

ALK fusion-hotspot benchmark report

Study: msk_impact_50k_2026

Abstract

This report analyzes ALK gene fusions from the msk_impact_50k_2026 study, covering 272 fusion events. 226/272 fusions (83.1%) were in-frame. 261/272 fusions (96.0%) retained the PF07714 domain. Domain retention was tested with Fisher's exact test ($p=0.00191661$, statistically significant at $\alpha=0.05$). No literature benchmark is configured for ALK.

1 Results summary

1.1 Fusion partner frequency

Fusion-partner frequency was tabulated across 272 analyzed fusion events, identifying 32 distinct fusion partner genes. The most frequent partner was EML4, observed in 221 events.

1.2 Domain retention

PF07714 domain retention was tested with Fisher's exact test comparing in-frame fusions against all others; 222/226 (98.2%) of in-frame fusions retained the domain, $p=0.00191661$ (statistically significant at $\alpha=0.05$). A breakpoint-position permutation test produced a corroborating empirical p-value of 0.00990099.

1.3 Domain disruption

Domain-disruption analysis was skipped: ALK has no disruption_required_domains configured; domain-disruption analysis was skipped.

1.4 Cutpoint detection

Cutpoint detection scanned 271 mapped breakpoints for the protein position that best separates domain-retained from lost/disrupted fusions; the inferred cutpoint was position 972 aa (permutation-corrected $p=0.00990099$, statistically significant at $\alpha=0.05$). This is 12 aa from the nearest configured domain boundary at 960 aa.

1.5 Corroborating confidence statistics

Corroborating confidence statistics were not computed for this run: ALK has no group_field configured; confidence-stats analysis was skipped.

1.6 Composite evidence score

Composite evidence score reported: domain_disruption sub-score excluded: the supplied result had no applicable p-value (e.g. disruption_required_domains not configured for this gene).

1.7 Exon retention

Exon retention reported: No target exon was supplied in params or GeneConfig.

1.8 Joint partner

Joint partner reported: ALK has no gene_pair configured; joint-partner analysis was skipped.

2 Tables

2.1 composite_score composite_evidence_ranking

Partner_gene	Event_count	Recurrence_score	Domain_retention_score	Domain_disruption_score	Cutpoint_proximity_score	Confidence_certainty_score	Components_applicable	Composite_score	Rank
EML4	221	0.8125	0.20043213737826426		0.8865878486549185		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.6097803023468625	1
LRP1B	1	0.003676470588235294	0.20043213737826426		1.0		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.2874442507443381	2
EMILIN1	1	0.003676470588235294	0.20043213737826426		0.9559412550066756		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.27800309110291144	3
CLTC	2	0.007352941176470588	0.20043213737826426		0.931909212283044		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.27442899791423403	4
FN1	1	0.003676470588235294	0.20043213737826426		0.931909212283044		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.2728533676621332	5
NOTCH1	1	0.003676470588235294	0.20043213737826426		0.931909212283044		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.2728533676621332	6
NRP2	1	0.003676470588235294	0.20043213737826426		0.931909212283044		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.2728533676621332	7
STRN	7	0.025735294117647058	0.20043213737826426		0.8865153538050734		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.2725798937866017	8
KIF5B	5	0.01838235294117647	0.20043213737826426		0.8945260347129506		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.27114520776265943	9
PLEKHA7	1	0.003676470588235294	0.20043213737826426		0.9158878504672897		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.2694202187016144	10
SPTBN1	4	0.014705882352941176	0.20043213737826426		0.8895193591455274		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.26849671846039647	11
RANBP2	3	0.011029411764705883	0.20043213737826426		0.8851802403204272		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.26599127703148845	12
PICALM	3	0.011029411764705883	0.20043213737826426		0.8811748998664887		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.26513298979135874	13
SQSTM1	2	0.007352941176470588	0.20043213737826426		0.8851802403204272		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.2644156467793876	14
TPM1	2	0.007352941176470588	0.20043213737826426		0.8851802403204272		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.2644156467793876	15
CTNNA1	1	0.003676470588235294	0.20043213737826426		0.8851802403204272		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.2628400165272868	16
DCTN1	1	0.003676470588235294	0.20043213737826426		0.8851802403204272		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.2628400165272868	17
DIAPH2	1	0.003676470588235294	0.20043213737826426		0.8851802403204272		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.2628400165272868	18
DYSF	1	0.003676470588235294	0.20043213737826426		0.8851802403204272		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.2628400165272868	19
GCC2	1	0.003676470588235294	0.20043213737826426		0.8851802403204272		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.2628400165272868	20
HMBOX1	1	0.003676470588235294	0.20043213737826426		0.8851802403204272		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.2628400165272868	21
PLEKHH2	1	0.003676470588235294	0.20043213737826426		0.8851802403204272		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.2628400165272868	22
PPP1CB	1	0.003676470588235294	0.20043213737826426		0.8851802403204272		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.2628400165272868	23
TFG	1	0.003676470588235294	0.20043213737826426		0.8851802403204272		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.2628400165272868	24

Partner_gene	Event_count	Recurrence_score	Domain_retention_score	Domain_disruption_score	Cutpoint_proximity_score	Confidence_certainty	Components_applicable	Composite_score	Rank
TRAF3	1	0.003676470588235294	0.20043213737826426		0.8851802403204272		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.2628400165272868	25
TTC27	1	0.003676470588235294	0.20043213737826426		0.8851802403204272		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.2628400165272868	26
VCL	1	0.003676470588235294	0.20043213737826426		0.8851802403204272		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.2628400165272868	27
ZFPM2	1	0.003676470588235294	0.20043213737826426		0.8851802403204272		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.2628400165272868	28
H6PD	1	0.003676470588235294	0.20043213737826426		0.7503337783711616		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.23394434610958698	29
CTSE	1	0.003676470588235294	0.20043213737826426		0.39118825100133514		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.1569845902446242	30
SOS1	1	0.003676470588235294	0.20043213737826426				['domain_retention', 'recurrence']	0.093110864583703	31
ASAP2	1	0.003676470588235294	0.20043213737826426		0.0		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.0731585364586238	32

2.2 cutpoint_detection cutpoint_scan

cutpoint	n_positive_at_or_below	n_positive_above	n_negative_at_or_below	n_negative_above	odds_ratio	p_value	neg_log10_p_value
223	0	261	1	9	0.0	0.03690036900369004	1.4329692908744058
516	0	261	2	8	0.0	0.001230012300123001	2.9100905455940684
785	0	261	3	7	0.0	3.658029145347215e-05	4.436752838604533
939	2	259	3	7	0.018018018018018018	0.0003515999093716071	3.4539512455419974
972	2	259	4	6	0.011583011583011582	1.3820564716971664e-05	4.8594742110534295
1023	10	251	4	6	0.05976095617529881	0.0007961806348620647	3.098988389769786
1028	11	250	4	6	0.066	0.0010657190672512412	2.9723572639982643
1032	12	249	4	6	0.07228915662650602	0.0013947454739763106	2.8555050392241372
1035	13	248	4	6	0.07862903225806452	0.0017901726849544938	2.747105073765603
1040	14	247	4	6	0.08502024291497975	0.002258991897970247	2.646085326703862
1042	16	245	4	6	0.09795918367346938	0.003444927806265043	2.4628198749665864
1045	18	243	4	6	0.11111111111111111	0.005008607520231548	2.300282998750876
1047	19	242	4	6	0.11776859504132231	0.005949364209574388	2.225529443519691
1051	20	241	4	6	0.12448132780082988	0.007005060123461933	2.154588132872754
1052	22	239	4	6	0.13807531380753138	0.00948736346530418	2.0228544610771384
1058	254	7	10	0	0.0	1.0	-0.0
1059	255	6	10	0	0.0	1.0	-0.0
1061	258	3	10	0	0.0	1.0	-0.0
1064	259	2	10	0	0.0	1.0	-0.0
1070	260	1	10	0	0.0	1.0	-0.0

2.3 domain_retention frame_domain_contingency_table

222	39
4	6

2.4 domain_retention domain_retention_descriptives

Domain_key	Domain_name	Quantitative_call_count	Fully_retained_count	Truncated_count	Fully_lost_count	Mean_retained_fraction_among_non_retained
kinase	Protein kinase domain	271	261	0	10	0.0

2.5 frequency Partner_gene_counts

Partner_gene	Event_count
ASAP2	1
CLTC	2
CTNNA1	1
CTSE	1
DCTN1	1
DIAPH2	1
DYSF	1
EMILIN1	1
EML4	221
FN1	1
GCC2	1
H6PD	1
HMBOX1	1
KIF5B	5
LRP1B	1
NOTCH1	1
NRP2	1
PICALM	3
PLEKHA7	1
PLEKHH2	1
PPP1CB	1
RANBP2	3
SOS1	1
SPTBN1	4
SQSTM1	2
STRN	7
TFG	1
TPM1	2
TRAF3	1
TTC27	1
VCL	1
ZFPM2	1

2.6 Per-event results (results.tsv)

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_1	breakpoint_2	breakpoint_3	protein_in_frame	intron_in_frame	domain_retention	mean_fraction_retained	fully_lost	disrupted_domains	annotation	effect_on_fusion
EVT-P-0024679-T01-IM6-39	P-0024679-T01-IM6		ALK-CTSE	CTSE	unknown	five_prime	29543594	7	516	True	lost	0.0	False		Protein kinase domain	ALK (NM_004304) - CTSE(NM_001910) Rear-range-ment: t(1;2)(q32.1;p23.3)(chr1:g.206318648; chr2:g.29543594) Antisense Fusion	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_1	breakpoint_2	breakpoint_3	protein_1	protein_2	breakpoint_4	is_main_ref	is_disrupted	is_disrupted	description	annotation	effect_on_fra
EVT-P-0020387-T01-IM6-40	P-0020387-T01-IM6		ALK-DCTN1	DCTN1	in-frame	three_prime	29446896	20	1058	True	retained	1.0	False	Protein kinase domain		DCTN1 (NM_004082) - ALK (NM_004304) rearrangement: c.3196+238:502:ALKdel Protein Fusion: in frame {DCTN1:ALK} ALK (NM_004304) - DYSF (NM_001130987) rearrangement: c.3172+909: Antisense fusion EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-13 fused to ALK exons 20-29): c.1490-639:EML4_c.3173-252:ALKinv Protein Fusion: in frame {EML4:ALK} EML4 (NM_001145076) - ALK (NM_004304) fusion (EML4 exons 1-5 fused with ALK exons 20-29): c.494-2623:EML4_c.3173-267:ALKinv Protein Fusion: in frame {EML4:ALK} EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-6 with ALK exons 20-29): c.668-6693:EML4_c.3173-257:ALKinv Protein fusion: in frame (EML4-ALK)	NA	
EVT-P-0006776-T01-IM5-44	P-0006776-T01-IM5		ALK-DYSF	DYSF	unknown	five_prime	29447418	19	1058	True	lost	0.0	False	Protein kinase domain		ALK (NM_004304) - DYSF (NM_001130987) rearrangement: c.3172+909: Antisense fusion EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-13 fused to ALK exons 20-29): c.1490-639:EML4_c.3173-252:ALKinv Protein Fusion: in frame {EML4:ALK} EML4 (NM_001145076) - ALK (NM_004304) fusion (EML4 exons 1-5 fused with ALK exons 20-29): c.494-2623:EML4_c.3173-267:ALKinv Protein Fusion: in frame {EML4:ALK} EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-6 with ALK exons 20-29): c.668-6693:EML4_c.3173-257:ALKinv Protein fusion: in frame (EML4-ALK)	NA	
EVT-P-0031827-T02-IM6-45	P-0031827-T02-IM6		ALK-EML4	EML4	in-frame	three_prime	29446646	20	1058	True	retained	1.0	False	Protein kinase domain		EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-13 fused to ALK exons 20-29): c.1490-639:EML4_c.3173-252:ALKinv Protein Fusion: in frame {EML4:ALK} EML4 (NM_001145076) - ALK (NM_004304) fusion (EML4 exons 1-5 fused with ALK exons 20-29): c.494-2623:EML4_c.3173-267:ALKinv Protein Fusion: in frame {EML4:ALK} EML4 (NM_001145076) - ALK (NM_004304) fusion (EML4 exons 1-6 with ALK exons 20-29): c.668-6693:EML4_c.3173-257:ALKinv Protein fusion: in frame (EML4-ALK)	NA	
EVT-P-0028287-T01-IM6-46	P-0028287-T01-IM6		ALK-EML4	EML4	in-frame	three_prime	29446661	20	1058	True	retained	1.0	False	Protein kinase domain		EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-5 fused with ALK exons 20-29): c.494-2623:EML4_c.3173-267:ALKinv Protein Fusion: in frame {EML4:ALK} EML4 (NM_001145076) - ALK (NM_004304) fusion (EML4 exons 1-6 with ALK exons 20-29): c.668-6693:EML4_c.3173-257:ALKinv Protein fusion: in frame (EML4-ALK)	NA	
EVT-P-0009301-T01-IM5-47	P-0009301-T01-IM5		ALK-EML4	EML4	in-frame	three_prime	29446651	20	1058	True	retained	1.0	False	Protein kinase domain		EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-6 with ALK exons 20-29): c.668-6693:EML4_c.3173-257:ALKinv Protein fusion: in frame (EML4-ALK)	NA	

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_pos	breakpoint_pos	breakpoint_pos	protein_pos	protein_pos	breakpoint_pos	breakpoint_pos	main_ref	donor_pos	acceptor_pos	disrupted_domain	description	annotation	effect_on_fra
EVT-P-0009301-T02-IM6-48	P-0009301-T02-IM6		ALK-EML4	EML4	in-frame	three_prime	29446651	20	1058	True	retained	1.0	False	Protein kinase domain				EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-5 fused to ALK exons 20-29): c.668-6693:EML4_257:ALKInv Protein fusion: in frame {EML4:ALK}	NA	
EVT-P-0009339-T01-IM5-49	P-0009339-T01-IM5		ALK-EML4	EML4	in-frame	three_prime	29447889	19	1058	True	retained	1.0	False	Protein kinase domain				EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-6 fused with ALK exons 20-29): c.668-6870:EML4_	c.3172+438:ALKInv	
EVT-P-0009339-T02-IM5-50	P-0009339-T02-IM5		ALK-EML4	EML4	in-frame	three_prime	29447889	19	1058	True	retained	1.0	False	Protein kinase domain				EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-6 fused with ALK exons 20-29): c.668-6870:EML4_	c.3172+438:ALKInv	
EVT-P-0009343-T01-IM5-51	P-0009343-T01-IM5		ALK-EML4	EML4	in-frame	three_prime	29446521	20	1058	True	retained	1.0	False	Protein kinase domain				EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-6 fused with ALK exons 20-29): c.668-1093:EML4_127:ALKInv Protein fusion: in frame (EML4-ALK)	NA	

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_1	breakpoint_2	breakpoint_3	protein_retained	protein_lost	breakpoint_4	main_retained	main_lost	disrupted_domain	disrupted_domain	disrupted_domain	description	annotation	effect_on_fusion
EVT-P-0009353-T01-IM5-52	P-0009353-T01-IM5		ALK-EML4	EML4	in-frame	three_prime	29446465	20	1058	True	retained	1.0	False	Protein kinase domain				EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-6 fused with ALK exons 20-29): c.667+2407:71:ALKinv Protein fusion: in frame (EML4-ALK)	NA	
EVT-P-0031406-T01-IM6-53	P-0031406-T01-IM6		ALK-EML4	EML4	in-frame	three_prime	29446425	20	1058	True	retained	1.0	False	Protein kinase domain				EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-5 with ALK exons 20-29): c.668-2889:EML4_31:ALKinv Protein Fusion: in frame {EML4:ALK}	NA	
EVT-P-0032170-T01-IM6-54	P-0032170-T01-IM6		ALK-EML4	EML4	in-frame	three_prime	29447695	19	1058	True	retained	1.0	False	Protein kinase domain				EML4 (NM_019063) - ALK (NM_004304) fusion: c.1489+1822:2:ALKinv Protein Fusion: in frame {EML4:ALK}	NA	
EVT-P-0010041-T01-IM5-55	P-0010041-T01-IM5		ALK-EML4	EML4	in-frame	three_prime	29447942	19	1058	True	retained	1.0	False	Protein kinase domain				EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-6 fuses with ALK exons 20-29): c.701-6977:EML4_31:ALKinv Protein fusion: in frame (EML4-ALK)	NA	
EVT-P-0008332-T01-IM5-56	P-0008332-T01-IM5		ALK-EML4	EML4	unknown	five_prime	29446912	20	1058	True	lost	0.0	False	Protein kinase domain				EML4 (NM_019063) - ALK (NM_004304) rearrangement: c.1489+828:518:ALKdup Antisense fusion	NA	

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_pos	breakpoint_ref	breakpoint_chr	protein_pos	protein_ref	main_retained	other_retained	protein_domain	disrupted_domains	annotation	effect_on_fra
EVT-P-0010177-T01-IM5-57	P-0010177-T01-IM5		ALK-EML4	EML4	in-frame	three_prime	29447358	20	1058	True	retained	1.0	False	Protein kinase domain		EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-12 fused in frame with ALK exons 20-29): c.1490-2649:EML4_c.3173-964:ALKInv Protein fusion: in frame (EML4-ALK)	NA
EVT-P-0007705-T01-IM5-58	P-0007705-T01-IM5		ALK-EML4	EML4	unknown	three_prime	29447417	19	1058	True	retained	1.0	False	Protein kinase domain		EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-13 with ALK exons 20-29) : c.1594:EML4_c.3172+910:ALKInv Protein fusion: mid-exon (EML4-ALK)	NA
EVT-P-0010978-T01-IM5-59	P-0010978-T01-IM5		ALK-EML4	EML4	out-of-frame	three_prime	29446414	20	1058	True	retained	1.0	False	Protein kinase domain		EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-18 fused with ALK exons 20-29): c.2154+2237:EML4_c.3173-20:ALKInv Protein fusion: out of frame (EML4-ALK)	NA
EVT-P-0010978-T02-IM5-60	P-0010978-T02-IM5		ALK-EML4	EML4	out-of-frame	three_prime	29446414	20	1058	True	retained	1.0	False	Protein kinase domain		EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-18 fused with ALK exons 20-29): c.2154+2237:EML4_c.3173-20:ALKInv Protein fusion: out of frame (EML4-ALK)	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_1	breakpoint_2	breakpoint_3	protein_1	protein_2	breakpoint_4	is_main_ref	is_disrupted	is_in_frame	is_in_domain	disrupted_desc	annotation	effect_on_fra
EVT-P-0011018-T01-IM5-61	P-0011018-T01-IM5		ALK-EML4	EML4	in-frame	three_prime	29448314	19	1058	True	retained	1.0	False	Protein kinase domain			EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-6 with ALK exons 20-29) Protein fusion: in frame (EML4-ALK)	NA	
EVT-P-0011203-T01-IM5-62	P-0011203-T01-IM5		ALK-EML4	EML4	in-frame	three_prime	29447105	20	1058	True	retained	1.0	False	Protein kinase domain			EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-6 fused with ALK exons 20-29) Protein fusion: in frame (EML4-ALK)	NA	
EVT-P-0011203-T02-IM5-63	P-0011203-T02-IM5		ALK-EML4	EML4	in-frame	three_prime	29447096	20	1058	True	retained	1.0	False	Protein kinase domain			EML4 (NM_019063) - ALK (NM_004304) fusion: c.668-6805:EML4_702:ALKinv Protein fusion: in frame (EML4-ALK)	NA	
EVT-P-0011203-T03-IM6-64	P-0011203-T03-IM6		ALK-EML4	EML4	in-frame	three_prime	29447055	20	1058	True	retained	1.0	False	Protein kinase domain			EML4 (NM_001145076) - ALK (NM_004304) fusion (EML4 exon 5 fused to ALK exon 20): c.494-6851:EML4_661:ALKinv Protein Fusion: in frame {EML4:ALK}	NA	
EVT-P-0012679-T01-IM5-65	P-0012679-T01-IM5		ALK-EML4	EML4	in-frame	three_prime	29447466	19	1058	True	retained	1.0	False	Protein kinase domain			EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-12 with ALK exons 20-29): c.1489+2812:EML4_c.3172+861:ALKinv	NA	

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint	breakpoint	breakpoint	protein	protein	breakpoint	main_ref	donor	fract	main_ref	disrupted	description	annotation	effect_on_fra
EVT-P-0012778-T01-IM5-66	P-0012778-T01-IM5		ALK-EML4	EML4	in-frame	three_prime	29447442	19	1058	True	retained	1.0	False	Protein kinase domain				EML4 (NM_019063) - ALK (NM_004304) fusion: c.667+1320: EML4_c.3172+885:ALKInv	NA	2+885:ALKInv
EVT-P-0012778-T02-IM6-67	P-0012778-T02-IM6		ALK-EML4	EML4	in-frame	three_prime	29447442	19	1058	True	retained	1.0	False	Protein kinase domain				EML4 (NM_001145076) - ALK (NM_004304) fusion (EML4 exons 1-5 fused with ALK exons 20-29): c.493+1320: EML4_c.3172+885:ALKInv	NA	2+885:ALKInv
EVT-P-0007211-T01-IM5-68	P-0007211-T01-IM5		ALK-EML4	EML4	in-frame	three_prime	29447489	19	1058	True	retained	1.0	False	Protein kinase domain				Fusion: in frame {EML4:ALK} EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-6 with ALK exons 20-29) : c.667+165: EML4_c.3172+838:ALKInv	NA	
EVT-P-0007125-T01-IM5-69	P-0007125-T01-IM5		ALK-EML4	EML4	unknown	three_prime	29446597	20	1058	True	retained	1.0	False	Protein kinase domain				EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-14 with ALK exons 20-29) : c.1328: EML4_c.3173-203:ALKInv	NA	
EVT-P-0006950-T02-IM6-70	P-0006950-T02-IM6		ALK-EML4	EML4	in-frame	three_prime	29447384	19	1058	True	retained	1.0	False	Protein kinase domain				EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-13 with ALK exons 20-29): c.1489+2302: EML4_c.3172+943:ALKInv	NA	

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_1	breakpoint_2	breakpoint_3	protein_1	protein_2	breakpoint_4	is_main_ref	is_disrupted	is_disrupted	description	annotation	effect_on_fra
EVT-P-0013869-T01-IM5-75	P-0013869-T01-IM5		ALK-EML4	EML4	in-frame	three_prime	29446712	20	1058	True	retained	1.0	False	Protein kinase domain		EML4 (NM_019063) - ALK (NM_004304) Fusion (EML4 exons 1 to 13 with exons 20-29 of ALK) : c.1490-187:EML4_c.3173-318:ALKInv Protein fusion: in frame (EML4-ALK) EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-6 fused in-frame to ALK exons 20-29) : c.667+2294: EML4_c.3173-285:ALKInv Protein fusion: in frame (EML4-ALK) EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-5 fused to ALK exons 20-29) : c.667+6128: EML4_c.3172+783:ALKInv	NA	
EVT-P-0014270-T01-IM6-76	P-0014270-T01-IM6		ALK-EML4	EML4	in-frame	three_prime	29446679	20	1058	True	retained	1.0	False	Protein kinase domain		EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-6 fused in-frame to ALK exons 20-29) : c.667+2294: EML4_c.3173-285:ALKInv Protein fusion: in frame (EML4-ALK) EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-5 fused to ALK exons 20-29) : c.667+6128: EML4_c.3172+783:ALKInv	NA	
EVT-P-0014310-T01-IM6-77	P-0014310-T01-IM6		ALK-EML4	EML4	in-frame	three_prime	29447544	19	1058	True	retained	1.0	False	Protein kinase domain		EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-5 fused to ALK exons 20-29) : c.667+6128: EML4_c.3172+783:ALKInv	NA	
EVT-P-0006370-T01-IM5-78	P-0006370-T01-IM5		ALK-EML4	EML4	in-frame	three_prime	29448124	19	1058	True	retained	1.0	False	Protein kinase domain		EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-13 with ALK exons 20-29) : c.1490-2147:EML4_c.3172+203:ALKInv	NA	

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_1	breakpoint_2	breakpoint_3	protein_1_pos	protein_2_pos	breakpoint_4	is_main_ref	is_coding	is_intron	is_disrupted	description	annotation	effect_on_fra
EVT-P-0015434-T02-IM6-86	P-0015434-T02-IM6		ALK-EML4	EML4	in-frame	three_prime	294448197	19	1058	True	retained	1.0	False	Protein kinase domain			EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exon 13 fused to ALK exon20): c.1489+63:EML4_c.3172+130:ALKInv Protein Fusion: in frame {EML4:ALK}	NA	
EVT-P-0005321-T01-IM5-87	P-0005321-T01-IM5		ALK-EML4	EML4	in-frame	three_prime	294448250	19	1058	True	retained	1.0	False	Protein kinase domain			EML4 (NM_001145076) - ALK (NM_004304) Fusion (EML4 exon 13 fused with ALK exon 20) : c.2068+156:EML4_c.3173-693:ALKInv Protein fusion: in frame (EML4-ALK)	NA	
EVT-P-0029971-T01-IM6-88	P-0029971-T01-IM6		ALK-EML4	EML4	in-frame	three_prime	294447807	19	1058	True	retained	1.0	False	Protein kinase domain			EML4 (NM_019063) - ALK (NM_004304) fusion: c.668-119:EML4_c.3172+520:ALKInv Protein Fusion: in frame {EML4:ALK}	NA	
EVT-P-0015921-T03-IM6-89	P-0015921-T03-IM6		ALK-EML4	EML4	in-frame	three_prime	294446809	20	1058	True	retained	1.0	False	Protein kinase domain			EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-6 fused with ALK ex-ons 20-29): c.667+516:EML4_c.3173-415:ALKInv Protein Fusion: in frame {EML4:ALK}	NA	
EVT-P-0015961-T01-IM6-90	P-0015961-T01-IM6		ALK-EML4	EML4	in-frame	three_prime	294446526	20	1058	True	retained	1.0	False	Protein kinase domain			EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exon 6fused to ALK exon 20): c.668-6666:EML4_c.3173-132:ALKInv Protein Fusion: in frame {EML4:ALK}	NA	

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_1	breakpoint_2	breakpoint_3	protein_retained	breakpoint_4	main_ref	donor_frame	inframe	disrupted_domain	disrupted_domain	description	annotation	effect_on_fra
EVT-P-0015961-T02-IM6-91	P-0015961-T02-IM6		ALK-EML4	EML4	in-frame	three_prime	29446526	20	1058	True	retained	1.0	False	Protein kinase domain			EML4 (NM_019063) - ALK fusion (EML4 exons 1-6 fused to ALK exons 20-29): c.668-6666:EML4_132:ALKInv Protein Fusion: in frame {EML4:ALK}	NA	
EVT-P-0015961-T03-IM6-92	P-0015961-T03-IM6		ALK-EML4	EML4	in-frame	three_prime	29446526	20	1058	True	retained	1.0	False	Protein kinase domain			EML4 (NM_019063) - ALK fusion (EML4 exons 1-6 fused to ALK exons 20-29): c.668-6666:EML4_132:ALKInv Protein Fusion: in frame {EML4:ALK}	NA	
EVT-P-0031827-T01-IM6-93	P-0031827-T01-IM6		ALK-EML4	EML4	in-frame	three_prime	29446700	20	1058	True	retained	1.0	False	Protein kinase domain			EML4 (NM_019063) - ALK fusion: c.1490-673:EML4_306:ALKInv Protein Fusion: in frame {EML4:ALK}	NA	
EVT-P-0027119-T01-IM6-94	P-0027119-T01-IM6		ALK-EML4	EML4	in-frame	three_prime	29446711	20	1058	True	retained	1.0	False	Protein kinase domain			EML4 (NM_019063) - ALK fusion (EML4 exons 1-12 fused to ALK exons 20-29): c.1490-837:EML4_317:ALKInv Protein Fusion: in frame {EML4:ALK}	NA	
EVT-P-0029750-T01-IM6-95	P-0029750-T01-IM6		ALK-EML4	EML4	in-frame	three_prime	29446660	20	1058	True	retained	1.0	False	Protein kinase domain			EML4 (NM_019063) - ALK fusion (EML4 exons 1-13 fused to ALK exons 20-29): c.1490-2574:EML4_266:ALKInv Protein Fusion: in frame {EML4:ALK}	NA	

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_1	breakpoint_2	breakpoint_3	protein_retained	breakpoint_4	main_ref	donor_frame	inframe	disrupted_domain	disrupted_domain	disrupted_domain	description	annotation	effect_on_fra
EVT-P-0029550-T01-IM6-96	P-0029550-T01-IM6		ALK-EML4	EML4	in-frame	three_prime	29447863	19	1058	True	retained	1.0	False	Protein kinase domain				EML4 (NM_001145076) - ALK (NM_004304) fusion (EML4 exons 1-5 fused with ALK exons 20-29): c.494-721:EML4_c.3172+464:ALK	NA	Inv
EVT-P-0016280-T01-IM6-97	P-0016280-T01-IM6		ALK-EML4	EML4	in-frame	three_prime	29447905	19	1058	True	retained	1.0	False	Protein kinase domain				EML4 (NM_001145076) - ALK (NM_004304) rearrangement: c.1316-2732:EML4_c.3172+422:ALK	NA	Inv
EVT-P-0016580-T01-IM6-98	P-0016580-T01-IM6		ALK-EML4	EML4	in-frame	three_prime	29447579	19	1058	True	retained	1.0	False	Protein kinase domain				EML4 (NM_001145076) - ALK (NM_004304) fusion (EML4 exons 1-13 fused to ALK exons 20-29): c.1489+2257:EML4_c.3172+748:ALK	NA	Inv
EVT-P-0005305-T01-IM5-99	P-0005305-T01-IM5		ALK-EML4	EML4	in-frame	three_prime	29447764	19	1058	True	retained	1.0	False	Protein kinase domain				EML4 (NM_001145076) - ALK (NM_004304) fusion (EML4 exons 1-6 with ALK exons 20-29): c.668-5757:EML4_c.3172+563:ALK	NA	Inv
EVT-P-0016960-T01-IM6-100	P-0016960-T01-IM6		ALK-EML4	EML4	in-frame	three_prime	29447838	19	1058	True	retained	1.0	False	Protein kinase domain				EML4 (NM_001145076) - ALK (NM_004304) fusion (EML4 exons 1-6 fused to ALK exons 20-29): c.667+7453:EML4_c.3172+489:ALK	NA	Inv

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_1	breakpoint_2	breakpoint_3	protein_retained	breakpoint_4	is_main_ref	is_disrupted	is_disrupted	is_disrupted	is_disrupted	is_disrupted	description	annotation	effect_on_fusion
EVT-P-0017592-T01-IM5-101	P-0017592-T01-IM5		ALK- EML4	EML4	in-frame	three_prime	29447548	19	1058	True	retained	1.0	False	Protein kinase domain				EML4 (NM_001145076) - ALK (NM_004304) Fusion (EML4 exon 1-5 fused with ALK exon 20-29): : c.494- 5963:EML4_ Protein Fusion: in frame {EML4:ALK} EML4 (NM_001145076) - ALK (NM_004304) fusion (EML4 exons 1-5 fused with ALK exon 20-29): c.1315+641: Protein Fusion: in frame {EML4:ALK} EML4 (NM_001145076) - ALK (NM_004304) fusion (EML4 exons 1-12 fused with ALK ex- ons 20-29): c.1315+641: Protein Fusion: in frame {EML4:ALK} EML4 (NM_001145076) - ALK (NM_004304) fusion (EML4 exon 20 fused with ALK exon 20) : c.2068+156: 693:ALKInv Protein fusion: in frame (EML4- ALK) EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-6 fused to ALK ex- ons 20-29): c.667+7306: 215:ALKInv Protein Fusion: in frame {EML4:ALK} EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-6 fused to ALK ex- ons 20-29): c.667+7306: 215:ALKInv Protein Fusion: in frame {EML4:ALK}	NA c.3172+779:ALKInv	
EVT-P-0029400-T01-IM6-102	P-0029400-T01-IM6		ALK- EML4	EML4	in-frame	three_prime	29447668	19	1058	True	retained	1.0	False	Protein kinase domain				EML4 (NM_001145076) - ALK (NM_004304) fusion (EML4 exons 1-12 fused with ALK ex- ons 20-29): c.1315+641: Protein Fusion: in frame {EML4:ALK} EML4 (NM_001145076) - ALK (NM_004304) fusion (EML4 exon 20 fused with ALK exon 20) : c.2068+156: 693:ALKInv Protein fusion: in frame (EML4- ALK) EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-6 fused to ALK ex- ons 20-29): c.667+7306: 215:ALKInv Protein Fusion: in frame {EML4:ALK} EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-6 fused to ALK ex- ons 20-29): c.667+7306: 215:ALKInv Protein Fusion: in frame {EML4:ALK}	NA EML4_c.3172+659:ALKInv	
EVT-P-0004218-T01-IM5-103	P-0004218-T01-IM5		ALK- EML4	EML4	in-frame	three_prime	29447087	20	1058	True	retained	1.0	False	Protein kinase domain				EML4 (NM_001145076) - ALK (NM_004304) fusion (EML4 exon 20 fused with ALK exon 20) : c.2068+156: 693:ALKInv Protein fusion: in frame (EML4- ALK) EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-6 fused to ALK ex- ons 20-29): c.667+7306: 215:ALKInv Protein Fusion: in frame {EML4:ALK} EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-6 fused to ALK ex- ons 20-29): c.667+7306: 215:ALKInv Protein Fusion: in frame {EML4:ALK}	NA EML4_c.3173-	
EVT-P-0017978-T01-IM6-104	P-0017978-T01-IM6		ALK- EML4	EML4	in-frame	three_prime	29446609	20	1058	True	retained	1.0	False	Protein kinase domain				EML4 (NM_001145076) - ALK (NM_004304) fusion (EML4 exons 1-6 fused to ALK ex- ons 20-29): c.667+7306: 215:ALKInv Protein Fusion: in frame {EML4:ALK} EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-6 fused to ALK ex- ons 20-29): c.667+7306: 215:ALKInv Protein Fusion: in frame {EML4:ALK}	NA EML4_c.3173-	
EVT-P-0017978-T02-IM6-105	P-0017978-T02-IM6		ALK- EML4	EML4	in-frame	three_prime	29446609	20	1058	True	retained	1.0	False	Protein kinase domain				EML4 (NM_001145076) - ALK (NM_004304) fusion (EML4 exons 1-6 fused to ALK ex- ons 20-29): c.667+7306: 215:ALKInv Protein Fusion: in frame {EML4:ALK}	NA EML4_c.3173-	

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_1	breakpoint_2	breakpoint_3	protein_retained	protein_retained	breakpoint_4	main_ref	donor_frame	main_ref	disrupted_domain	disrupted_domain	description	annotation	effect_on_fra
EVT-P-0019491-T01-IM6-113	P-0019491-T01-IM6		ALK-EML4	EML4	in-frame	three_prime	29446596	20	1058	True	retained	1.0	False	Protein kinase domain				EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-12 with ALK exons 20-29): c.1490-486:EML4_c.3173-202:ALKInv Protein Fusion: in frame {EML4:ALK} EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-12 fused with ALK exons 19-29): c.1489+2278:EML4_c.3096:ALKInv Protein Fusion: mid-exon {EML4:ALK} EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-6 with ALK exons 20-29): c.667+6837:EML4_c.3173-100:ALKInv Protein Fusion: in frame {EML4:ALK} EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-12 fused in frame with ALK exons 20-29, which include the ALK kinase domain): c.1490-244:EML4_c.3173-119:ALKInv Protein fusion: in frame (EML4-ALK)	NA	
EVT-P-0019513-T01-IM6-114	P-0019513-T01-IM6		ALK-EML4	EML4	unknown	three_prime	29448403	19	1032	False	retained	1.0	False	Protein kinase domain				EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-12 fused with ALK exons 19-29): c.1489+2278:EML4_c.3096:ALKInv Protein Fusion: mid-exon {EML4:ALK} EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-6 with ALK exons 20-29): c.667+6837:EML4_c.3173-100:ALKInv Protein Fusion: in frame {EML4:ALK} EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-12 fused in frame with ALK exons 20-29, which include the ALK kinase domain): c.1490-244:EML4_c.3173-119:ALKInv Protein fusion: in frame (EML4-ALK)	NA	
EVT-P-0019533-T01-IM6-115	P-0019533-T01-IM6		ALK-EML4	EML4	in-frame	three_prime	29446494	20	1058	True	retained	1.0	False	Protein kinase domain				EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-12 fused with ALK exons 19-29): c.1489+2278:EML4_c.3096:ALKInv Protein Fusion: mid-exon {EML4:ALK} EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-6 with ALK exons 20-29): c.667+6837:EML4_c.3173-100:ALKInv Protein Fusion: in frame {EML4:ALK} EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-12 fused in frame with ALK exons 20-29, which include the ALK kinase domain): c.1490-244:EML4_c.3173-119:ALKInv Protein fusion: in frame (EML4-ALK)	NA	
EVT-P-0009102-T01-IM5-116	P-0009102-T01-IM5		ALK-EML4	EML4	in-frame	three_prime	29446513	20	1058	True	retained	1.0	False	Protein kinase domain				EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-12 fused in frame with ALK exons 20-29, which include the ALK kinase domain): c.1490-244:EML4_c.3173-119:ALKInv Protein fusion: in frame (EML4-ALK)	NA	

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_1	breakpoint_2	breakpoint_3	protein_retained	protein_retained	breakpoint_4	main_ref	donor_frame	main_ref	disrupted_domain	disrupted_domain	description	annotation	effect_on_fra
EVT-P-0028901-T01-IM6-117	P-0028901-T01-IM6		ALK-EML4	EML4	in-frame	three_prime	29446607	20	1058	True	retained	1.0	False	Protein kinase domain				EML4 (NM_001145076) - ALK (NM_004304) fusion (EML4 exons 1-12 fused with ALK exons 20-29): c.1316-844:EML4_c.3173-213:ALKInv Protein Fusion: in frame {EML4:ALK} EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-12 fused in-frame with ALK exons 20-29): c.2056+82:EML4_c.3172+752:ALKInv Protein Fusion: in frame {EML4:ALK} EML4 (NM_001145076) - ALK (NM_004304) fusion (EML4 exons 1-5 fused to ALK exon 20-29) : c.493+1179:EML4_c.3172+226:ALKInv Protein Fusion: in frame {EML4:ALK} EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-13 fused in-frame with ALK exons 20-29): c.1489+879:EML4_c.3172+372:ALKInv Protein Fusion: in frame {EML4:ALK} EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-6 fused to ALK exons 20-29): c.668-3016:EML4_c.3172+420:ALKInv Protein Fusion: in frame {EML4:ALK}	NA	
EVT-P-0019786-T01-IM6-118	P-0019786-T01-IM6		ALK-EML4	EML4	in-frame	three_prime	29447575	19	1058	True	retained	1.0	False	Protein kinase domain				EML4 (NM_001145076) - ALK (NM_004304) fusion (EML4 exons 1-18 fused in-frame with ALK exons 20-29): c.2056+82:EML4_c.3172+752:ALKInv Protein Fusion: in frame {EML4:ALK} EML4 (NM_001145076) - ALK (NM_004304) fusion (EML4 exons 1-5 fused to ALK exon 20-29) : c.493+1179:EML4_c.3172+226:ALKInv Protein Fusion: in frame {EML4:ALK} EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-13 fused in-frame with ALK exons 20-29): c.1489+879:EML4_c.3172+372:ALKInv Protein Fusion: in frame {EML4:ALK} EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-6 fused to ALK exons 20-29): c.668-3016:EML4_c.3172+420:ALKInv Protein Fusion: in frame {EML4:ALK}	NA	
EVT-P-0019972-T01-IM6-119	P-0019972-T01-IM6		ALK-EML4	EML4	in-frame	three_prime	29448101	19	1058	True	retained	1.0	False	Protein kinase domain				EML4 (NM_001145076) - ALK (NM_004304) fusion (EML4 exons 1-5 fused to ALK exon 20-29) : c.493+1179:EML4_c.3172+226:ALKInv Protein Fusion: in frame {EML4:ALK} EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-13 fused in-frame with ALK exons 20-29): c.1489+879:EML4_c.3172+372:ALKInv Protein Fusion: in frame {EML4:ALK} EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-6 fused to ALK exons 20-29): c.668-3016:EML4_c.3172+420:ALKInv Protein Fusion: in frame {EML4:ALK}	NA	
EVT-P-0020124-T01-IM6-120	P-0020124-T01-IM6		ALK-EML4	EML4	in-frame	three_prime	29447955	19	1058	True	retained	1.0	False	Protein kinase domain				EML4 (NM_001145076) - ALK (NM_004304) fusion (EML4 exons 1-13 fused in-frame with ALK exons 20-29): c.1489+879:EML4_c.3172+372:ALKInv Protein Fusion: in frame {EML4:ALK} EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-6 fused to ALK exons 20-29): c.668-3016:EML4_c.3172+420:ALKInv Protein Fusion: in frame {EML4:ALK}	NA	
EVT-P-0020138-T01-IM6-121	P-0020138-T01-IM6		ALK-EML4	EML4	in-frame	three_prime	29447907	19	1058	True	retained	1.0	False	Protein kinase domain				EML4 (NM_001145076) - ALK (NM_004304) fusion (EML4 exons 1-6 fused to ALK exons 20-29): c.668-3016:EML4_c.3172+420:ALKInv Protein Fusion: in frame {EML4:ALK}	NA	

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_1	breakpoint_2	breakpoint_3	protein_retained	breakpoint_4	main_ref	donor_frame	maintained	domain	disrupted_domain	description	annotation	effect_on_fra
EVT-P-0003383-T01-IM5-122	P-0003383-T01-IM5		ALK- EML4	EML4	in-frame	three_prime	29447572	19	1058	True	retained	1.0	False	Protein kinase domain			NA Protein fusion: in frame (EML4-ALK)	NA	
EVT-P-0020408-T01-IM6-123	P-0020408-T01-IM6		ALK- EML4	EML4	in-frame	three_prime	29446779	20	1058	True	retained	1.0	False	Protein kinase domain			EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-13 fused to ALK exons 20-29): c.1489+1227:385:ALKInv Protein Fusion: in frame {EML4:ALK}	NA	EML4_c.3173-
EVT-P-0003323-T01-IM5-124	P-0003323-T01-IM5		ALK- EML4	EML4	in-frame	three_prime	29447530	19	1058	True	retained	1.0	False	Protein kinase domain			NA Protein fusion: in frame (EML4-ALK)	NA	
EVT-P-0003297-T01-IM5-125	P-0003297-T01-IM5		ALK- EML4	EML4	in-frame	three_prime	29448259	19	1058	True	retained	1.0	False	Protein kinase domain			NA Protein fusion: in frame (EML4-ALK)	NA	
EVT-P-0020584-T01-IM6-126	P-0020584-T01-IM6		ALK- EML4	EML4	in-frame	three_prime	29447048	20	1058	True	retained	1.0	False	Protein kinase domain			EML4 (NM_001145076) - ALK (NM_004304) fusion (EML4 exons 1-2 fused to ALK exon 20-29): c.209-4846:EML4_654:ALKInv Protein Fusion: in frame {EML4:ALK}	NA	EML4_c.3173-
EVT-P-0020780-T01-IM6-127	P-0020780-T01-IM6		ALK- EML4	EML4	in-frame	three_prime	29447084	20	1058	True	retained	1.0	False	Protein kinase domain			EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1 - 13 fused to ALK exons 20 - 29): c.1489+1278:690:ALKInv Protein Fusion: in frame {EML4:ALK}	NA	EML4_c.3173-
EVT-P-0020825-T03-IM6-128	P-0020825-T03-IM6		ALK- EML4	EML4	in-frame	three_prime	29446767	20	1058	True	retained	1.0	False	Protein kinase domain			EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-2 fused to ALK exons 20-29):c.208+4373:ALKInv Protein Fusion: in frame {EML4:ALK}	NA	EML4_c.3173-

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_pos	breakpoint_pos	breakpoint_pos	protein_pos	breakpoint_pos	main_ref	chr	start	end	disrupted_domain	description	annotation	effect_on_fra
EVT-P-0021257-T01-IM6-129	P-0021257-T01-IM6		ALK-EML4	EML4	in-frame	three_prime	29447777	19	1058	True	retained	1.0	False	Protein kinase domain			EML4 (NM_001145076) - ALK (NM_004304) Fusion (EML4 exons 1-17 fused with ALK exons 20-29): c.1883-596:EML4_c.3172+550:ALKInv	NA	
EVT-P-0003206-T01-IM5-130	P-0003206-T01-IM5		ALK-EML4	EML4	unknown	three_prime	29448365	19	1045	False	retained	1.0	False	Protein kinase domain			Fusion: in frame {EML4:ALK} NA NA Protein fusion: mid-exon (EML4-ALK) EML4 (NM_001145076) - ALK (NM_004304) Fusion (EML4 exons 1-5 fused with ALK exons 20-29): c.493+2355:EML4_c.3172+14:ALKInv	NA	
EVT-P-0021495-T01-IM6-131	P-0021495-T01-IM6		ALK-EML4	EML4	in-frame	three_prime	29448313	19	1058	True	retained	1.0	False	Protein kinase domain			EML4 (NM_001145076) - ALK (NM_004304) Fusion (EML4 exons 1-5 fused with ALK exons 20-29): c.493+2355:EML4_c.3172+14:ALKInv	NA	
EVT-P-0002613-T01-IM3-132	P-0002613-T01-IM3		ALK-EML4	EML4	in-frame	three_prime	29447199	20	1058	True	retained	1.0	False	Protein kinase domain			Fusion: in frame {EML4:ALK} NA NA Protein fusion: in frame (EML4-ALK) EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-18 fuses with ALK exons 20-29): c.2057-115:EML4_c.3172+274:ALKInv	NA	
EVT-P-0002408-T02-IM5-133	P-0002408-T02-IM5		ALK-EML4	EML4	in-frame	three_prime	29448053	19	1058	True	retained	1.0	False	Protein kinase domain			Fusion (EML4 exons 1-14 fused with ALK exons 20-29): c.1316-1509:EML4_c.3173-608:ALKInv	NA	
EVT-P-0021693-T01-IM6-134	P-0021693-T01-IM6		ALK-EML4	EML4	in-frame	three_prime	29447002	20	1058	True	retained	1.0	False	Protein kinase domain			Fusion (EML4 exons 1-14 fused with ALK exons 20-29): c.1316-1509:EML4_c.3173-608:ALKInv	NA	

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_1	breakpoint_2	breakpoint_3	protein_retained	breakpoint_4	main_ref	clonally_expressed	in_frame	disrupted_domain	disrupted_domain	description	annotation	effect_on_fra
EVT-P-0022101-T01-IM6-135	P-0022101-T01-IM6		ALK-EML4	EML4	in-frame	three_prime	29447759	19	1058	True	retained	1.0	False	Protein kinase domain			EML4 (NM_019063) - ALK (NM_004304) Fusion (EML4 exon18 fused with ALK exon 20): c.2057-510:EML4_c.2057-510:ALKInv Protein Fusion: in frame {EML4:ALK} EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-20 fused with ALK exons 20-29): c.2333:EML4_c.3173-190:ALKInv Protein Fusion: mid-exon {EML4:ALK} EML4 (NM_019063) - ALK (NM_004304) fusion: c.1489+107:EML4_c.3172+470:ALKInv Protein Fusion: in frame {EML4:ALK} EML4 (NM_019063) -ALK (NM_004304) fusion (EML4 exons 1-13 fused to ALK exons 20-29): c.1490-2055:EML4_c.3173-536:ALKInv Protein Fusion: in frame {EML4:ALK} EML4 (NM_001145076) - ALK (NM_004304) fusion (EML4 exons 1-12 fused with ALK exons 20-29): c.1523-2563:EML4_c.3172:ALKInv Protein Fusion: in frame {EML4:ALK}	NA	
EVT-P-0022138-T01-IM6-136	P-0022138-T01-IM6		ALK-EML4	EML4	unknown	three_prime	29446584	20	1058	True	retained	1.0	False	Protein kinase domain			EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-20 fused with ALK exons 20-29): c.2333:EML4_c.3173-190:ALKInv Protein Fusion: mid-exon {EML4:ALK} EML4 (NM_019063) - ALK (NM_004304) fusion: c.1489+107:EML4_c.3172+470:ALKInv Protein Fusion: in frame {EML4:ALK} EML4 (NM_019063) -ALK (NM_004304) fusion (EML4 exons 1-13 fused to ALK exons 20-29): c.1490-2055:EML4_c.3173-536:ALKInv Protein Fusion: in frame {EML4:ALK} EML4 (NM_001145076) - ALK (NM_004304) fusion (EML4 exons 1-12 fused with ALK exons 20-29): c.1523-2563:EML4_c.3172:ALKInv Protein Fusion: in frame {EML4:ALK}	NA	
EVT-P-0036247-T01-IM6-137	P-0036247-T01-IM6		ALK-EML4	EML4	in-frame	three_prime	29447857	19	1058	True	retained	1.0	False	Protein kinase domain			EML4 (NM_019063) - ALK (NM_004304) fusion: c.1489+107:EML4_c.3172+470:ALKInv Protein Fusion: in frame {EML4:ALK} EML4 (NM_019063) -ALK (NM_004304) fusion (EML4 exons 1-13 fused to ALK exons 20-29): c.1490-2055:EML4_c.3173-536:ALKInv Protein Fusion: in frame {EML4:ALK} EML4 (NM_001145076) - ALK (NM_004304) fusion (EML4 exons 1-12 fused with ALK exons 20-29): c.1523-2563:EML4_c.3172:ALKInv Protein Fusion: in frame {EML4:ALK}	NA	
EVT-P-0036106-T01-IM6-138	P-0036106-T01-IM6		ALK-EML4	EML4	in-frame	three_prime	29446930	20	1058	True	retained	1.0	False	Protein kinase domain			EML4 (NM_019063) -ALK (NM_004304) fusion (EML4 exons 1-13 fused to ALK exons 20-29): c.1490-2055:EML4_c.3173-536:ALKInv Protein Fusion: in frame {EML4:ALK} EML4 (NM_001145076) - ALK (NM_004304) fusion (EML4 exons 1-12 fused with ALK exons 20-29): c.1523-2563:EML4_c.3172:ALKInv Protein Fusion: in frame {EML4:ALK}	NA	
EVT-P-0028745-T01-IM6-139	P-0028745-T01-IM6		ALK-EML4	EML4	in-frame	three_prime	29447509	19	1058	True	retained	1.0	False	Protein kinase domain			EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-12 fused with ALK exons 20-29): c.1523-2563:EML4_c.3172:ALKInv Protein Fusion: in frame {EML4:ALK}	NA	

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_1	breakpoint_2	breakpoint_3	protein_positives	biomarker	main_retention	dominant_fusion	disrupted_domains	disrupted_domains	disrupted_domains	disrupted_domains	disrupted_domains	
EVT-P-0022398-T01-IM6-140	P-0022398-T01-IM6		ALK-EML4	EML4	in-frame	three_prime	29448035	19	1058	True	retained	1.0	False	Protein kinase domain			EML4 (NM_001145076) - ALK (NM_004304) fusion (EML4 exons 1-5 fused with ALK exons 20-29): c.494-7853:EML4_c.3172+292:ALKInv Protein Fusion: in frame {EML4:ALK} NA (NM_004304) and EML4 (NM_019063) reciprocal inversion (13026566bp): c.3173-889:ALK_c.208+1022:EML4inv	NA	Effect on function
EVT-P-0002266-T01-IM3-141	P-0002266-T01-IM3		ALK-EML4	EML4	in-frame	five_prime	29447288	20	1058	True	lost	0.0	False		Protein kinase domain		EML4 (NM_001145076) - ALK (NM_004304) and EML4 (NM_019063) reciprocal inversion (13026566bp): c.3173-889:ALK_c.208+1022:EML4inv	NA	Effect on function
EVT-P-0002266-T01-IM3-142	P-0002266-T01-IM3		ALK-EML4	EML4	in-frame	three_prime	29447283	20	1058	True	retained	1.0	False	Protein kinase domain			NA NA	Effect on function	
EVT-P-0028541-T03-IM6-143	P-0028541-T03-IM6		ALK-EML4	EML4	in-frame	three_prime	29446922	20	1058	True	retained	1.0	False	Protein kinase domain			EML4 (NM_001145076) - ALK (NM_004304) fusion: c.208+4631:EML4_c.3173-528:ALKInv Protein Fusion: in frame {EML4:ALK} NA (NM_001145076) - ALK (NM_004304) fusion (EML4 exons 1-5 fused with ALK exons 20-29): c.493+983:EML4_c.3173-164:ALKInv Protein Fusion: in frame {EML4:ALK} NA (NM_019063) - ALK (NM_004304) fusion: c.668-4857:EML4_c.3172+944:ALKInv	NA	Effect on function
EVT-P-0022926-T01-IM6-144	P-0022926-T01-IM6		ALK-EML4	EML4	in-frame	three_prime	29446558	20	1058	True	retained	1.0	False	Protein kinase domain			EML4 (NM_001145076) - ALK (NM_004304) fusion (EML4 exons 1-5 fused with ALK exons 20-29): c.493+983:EML4_c.3173-164:ALKInv Protein Fusion: in frame {EML4:ALK} NA (NM_001145076) - ALK (NM_004304) fusion: c.668-4857:EML4_c.3172+944:ALKInv	NA	Effect on function
EVT-P-0035121-T01-IM6-145	P-0035121-T01-IM6		ALK-EML4	EML4	in-frame	three_prime	29447383	19	1058	True	retained	1.0	False	Protein kinase domain			EML4 (NM_001145076) - ALK (NM_004304) fusion: c.668-4857:EML4_c.3172+944:ALKInv	NA	Effect on function
EVT-P-0002217-T02-IM3-146	P-0002217-T02-IM3		ALK-EML4	EML4	in-frame	three_prime	29446594	20	1058	True	retained	1.0	False	Protein kinase domain			NA NA	Effect on function	

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_5p	breakpoint_3p	breakpoint_5p	breakpoint_3p	protein_retained	protein_retained	main_ref	donor_frame	main_ref	disrupted_domain	disrupted_domain	description_annotation	status	effect_on_fusion
EVT-P-0023368-T01-IM6-147	P-0023368-T01-IM6		ALK- EML4	EML4	in-frame	three_prime	29446713	20	1058	True	retained	1.0	False	Protein kinase domain				EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-5 fused with ALK exons 20-29 including the kinase domain) : c.667+7004: EML4_c.3173-319:ALKInv Protein Fusion: in frame {EML4:ALK} EML4 (NM_019063) - ALK (NM_004304) inversion: c.2242+31: EML4_c.3173-739:ALKInv Protein fusion: in frame (EML4-ALK) EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-5 fused with ALK exons 19-29): c.667+4775: EML4_c.3068-389:ALKInv Protein Fusion: in frame {EML4:ALK} EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-6 fused to ALK exons 20 -29): c.668-4498: EML4_c.3172+61: ALKInv Protein Fusion: in frame {EML4:ALK} EML4 (NM_019063) - ALK (NM_004304) Inversion: c.1315+2666: EML4_c.3172+122: ALKInv Protein fusion: in frame (EML4-ALK)	NA	
EVT-P-0002024-T01-IM3-148	P-0002024-T01-IM3		ALK- EML4	EML4	in-frame	three_prime	29447133	20	1058	True	retained	1.0	False	Protein kinase domain				EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-5 fused with ALK exons 20-29 including the kinase domain) : c.667+7004: EML4_c.3173-319:ALKInv Protein Fusion: in frame {EML4:ALK} EML4 (NM_019063) - ALK (NM_004304) inversion: c.2242+31: EML4_c.3173-739:ALKInv Protein fusion: in frame (EML4-ALK) EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-5 fused with ALK exons 19-29): c.667+4775: EML4_c.3068-389:ALKInv Protein Fusion: in frame {EML4:ALK} EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-6 fused to ALK exons 20 -29): c.668-4498: EML4_c.3172+61: ALKInv Protein Fusion: in frame {EML4:ALK} EML4 (NM_019063) - ALK (NM_004304) Inversion: c.1315+2666: EML4_c.3172+122: ALKInv Protein fusion: in frame (EML4-ALK)	NA	
EVT-P-0023962-T01-IM6-149	P-0023962-T01-IM6		ALK- EML4	EML4	in-frame	three_prime	29448820	19	1023	True	retained	1.0	False	Protein kinase domain				EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-5 fused with ALK exons 19-29): c.667+4775: EML4_c.3068-389:ALKInv Protein Fusion: in frame {EML4:ALK} EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-6 fused to ALK exons 20 -29): c.668-4498: EML4_c.3172+61: ALKInv Protein Fusion: in frame {EML4:ALK} EML4 (NM_019063) - ALK (NM_004304) Inversion: c.1315+2666: EML4_c.3172+122: ALKInv Protein fusion: in frame (EML4-ALK)	NA	
EVT-P-0024136-T01-IM6-150	P-0024136-T01-IM6		ALK- EML4	EML4	in-frame	three_prime	29448266	19	1058	True	retained	1.0	False	Protein kinase domain				EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-6 fused to ALK exons 20 -29): c.668-4498: EML4_c.3172+61: ALKInv Protein Fusion: in frame {EML4:ALK} EML4 (NM_019063) - ALK (NM_004304) Inversion: c.1315+2666: EML4_c.3172+122: ALKInv Protein fusion: in frame (EML4-ALK)	NA	
EVT-P-0001811-T01-IM3-151	P-0001811-T01-IM3		ALK- EML4	EML4	in-frame	three_prime	29448205	19	1058	True	retained	1.0	False	Protein kinase domain				EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-6 fused to ALK exons 20 -29): c.668-4498: EML4_c.3172+61: ALKInv Protein Fusion: in frame {EML4:ALK} EML4 (NM_019063) - ALK (NM_004304) Inversion: c.1315+2666: EML4_c.3172+122: ALKInv Protein fusion: in frame (EML4-ALK)	NA	

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_pos	breakpoint_pos	breakpoint_pos	protein_pos	breakpoint_pos	main_ref	donor_pos	acceptor_pos	disrupted_domain	description	annotation	effect_on_fra
EVT-P-0028541-T02-IM6-152	P-0028541-T02-IM6		ALK- EML4	EML4	in-frame	three_prime	29446922	20	1058	True	retained	1.0	False	Protein kinase domain		EML4 (NM_001145076) - ALK (NM_004304) fusion (EML4 exons 1-2 fused with ALK ex- ons 20-29): c.208+4631- 528:ALKInv Protein Fusion: in frame {EML4:ALK} EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-2 fused to ALK exons 20 -29): c.209- 990:EML4_ 483:ALKInv Protein Fusion: in frame {EML4:ALK} EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1- 20 fused to ALK exons 20-29): c.2243- 217:EML4_ 120:ALKInv Protein Fusion: in frame {EML4:ALK} EML4 (NM_019063) - ALK (NM_004304) Fusion (EML4 exon13 fused with ALK ex- ons 20) : c.1641+555- Fusion: mid-exon {EML4:ALK} NA NA Protein fusion: in frame (EML4- ALK)	NA 5076) 4) - EML4_c.3173-	
EVT-P-0024907-T01-IM6-153	P-0024907-T01-IM6		ALK- EML4	EML4	in-frame	three_prime	29446877	20	1058	True	retained	1.0	False	Protein kinase domain		EML4 (NM_001145076) - ALK (NM_004304) fusion (EML4 exons 1-2 fused to ALK exons 20 -29): c.209- 990:EML4_ 483:ALKInv Protein Fusion: in frame {EML4:ALK} EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1- 20 fused to ALK exons 20-29): c.2243- 217:EML4_ 120:ALKInv Protein Fusion: in frame {EML4:ALK} EML4 (NM_019063) - ALK (NM_004304) Fusion (EML4 exon13 fused with ALK ex- ons 20) : c.1641+555- Fusion: mid-exon {EML4:ALK} NA NA Protein fusion: in frame (EML4- ALK)	NA 5076) 4) - EML4_c.3173-	
EVT-P-0034569-T02-IM6-154	P-0034569-T02-IM6		ALK- EML4	EML4	in-frame	three_prime	29446514	20	1058	True	retained	1.0	False	Protein kinase domain		EML4 (NM_001145076) - ALK (NM_004304) fusion (EML4 exons 1-2 fused to ALK exons 20 -29): c.209- 990:EML4_ 483:ALKInv Protein Fusion: in frame {EML4:ALK} EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1- 20 fused to ALK exons 20-29): c.2243- 217:EML4_ 120:ALKInv Protein Fusion: in frame {EML4:ALK} EML4 (NM_019063) - ALK (NM_004304) Fusion (EML4 exon13 fused with ALK ex- ons 20) : c.1641+555- Fusion: mid-exon {EML4:ALK} NA NA Protein fusion: in frame (EML4- ALK)	NA 5076) 4) - EML4_c.3173-	
EVT-P-0034450-T01-IM6-155	P-0034450-T01-IM6		ALK- EML4	EML4	unknown	three_prime	29446354	20	1071	False	retained	1.0	False	Protein kinase domain		EML4 (NM_001145076) - ALK (NM_004304) fusion (EML4 exons 1-2 fused to ALK exons 20 -29): c.209- 990:EML4_ 483:ALKInv Protein Fusion: in frame {EML4:ALK} EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1- 20 fused to ALK exons 20-29): c.2243- 217:EML4_ 120:ALKInv Protein Fusion: in frame {EML4:ALK} EML4 (NM_019063) - ALK (NM_004304) Fusion (EML4 exon13 fused with ALK ex- ons 20) : c.1641+555- Fusion: mid-exon {EML4:ALK} NA NA Protein fusion: in frame (EML4- ALK)	NA 5076) 4) - EML4_c.3173-	
EVT-P-0001532-T01-IM3-156	P-0001532-T01-IM3		ALK- EML4	EML4	in-frame	three_prime	29447553	19	1058	True	retained	1.0	False	Protein kinase domain		EML4 (NM_001145076) - ALK (NM_004304) fusion (EML4 exons 1-2 fused to ALK exons 20 -29): c.209- 990:EML4_ 483:ALKInv Protein Fusion: in frame {EML4:ALK} EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1- 20 fused to ALK exons 20-29): c.2243- 217:EML4_ 120:ALKInv Protein Fusion: in frame {EML4:ALK} EML4 (NM_019063) - ALK (NM_004304) Fusion (EML4 exon13 fused with ALK ex- ons 20) : c.1641+555- Fusion: mid-exon {EML4:ALK} NA NA Protein fusion: in frame (EML4- ALK)	NA 5076) 4) - EML4_c.3173-	

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_1	breakpoint_2	breakpoint_3	protein_coding	in_frame	main_retained	donor_insertion	maintained_domain	disrupted_domains	annotation	effect_on_fusion
EVT-P-0028541-T01-IM6-157	P-0028541-T01-IM6		ALK-EML4	EML4	in-frame	three_prime	29446922	20	1058	True	retained	1.0	False	Protein kinase domain		EML4 (NM_001145076) - ALK (NM_004304) fusion (EML4 exons 1-2 fused with ALK exons 20-29): c.208+4631-528:ALKInv Protein Fusion: in frame {EML4:ALK} EML4 NA (NM_001145076) - ALK (NM_004304) fusion (EML4 exons 1-18 fused with ALK exons 20-29): c.1981-2450:EML4_c.3173:ALKInv Protein Fusion: mid-exon {EML4:ALK} EML4 NA (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-5 fuses with ALK exons 20-29) : c.667+7374:EML4_c.3172+876:ALKInv Protein Fusion: in frame {ALK:EML4} EML4 NA (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-20 fuses with ALK exons 20-29) : c.2312:EML4_c.3173-558:ALKInv Protein Fusion: mid-exon {EML4:ALK}	
EVT-P-0026407-T01-IM6-158	P-0026407-T01-IM6		ALK-EML4	EML4	unknown	three_prime	29446392	20	1059	False	retained	1.0	False	Protein kinase domain		EML4 NA (NM_001145076) - ALK (NM_004304) fusion (EML4 exons 1-18 fused with ALK exons 20-29): c.1981-2450:EML4_c.3175:ALKInv Protein Fusion: mid-exon {EML4:ALK} EML4 NA (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-5 fuses with ALK exons 20-29) : c.667+7374:EML4_c.3172+876:ALKInv Protein Fusion: in frame {ALK:EML4} EML4 NA (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-20 fuses with ALK exons 20-29) : c.2312:EML4_c.3173-558:ALKInv Protein Fusion: mid-exon {EML4:ALK}	
EVT-P-0026650-T02-IM6-159	P-0026650-T02-IM6		ALK-EML4	EML4	in-frame	five_prime	29447451	19	1058	True	lost	0.0	False	Protein kinase domain		EML4 NA (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-5 fuses with ALK exons 20-29) : c.667+7374:EML4_c.3172+876:ALKInv Protein Fusion: in frame {ALK:EML4} EML4 NA (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-20 fuses with ALK exons 20-29) : c.2312:EML4_c.3173-558:ALKInv Protein Fusion: mid-exon {EML4:ALK}	
EVT-P-0026657-T01-IM6-160	P-0026657-T01-IM6		ALK-EML4	EML4	unknown	three_prime	29446952	20	1058	True	retained	1.0	False	Protein kinase domain		EML4 NA (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-20 fuses with ALK exons 20-29) : c.2312:EML4_c.3173-558:ALKInv Protein Fusion: mid-exon {EML4:ALK} EML4 NA (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-5 fused to ALK exons 20-29): c.2312:EML4_c.3173-558:ALKInv Protein Fusion: mid-exon {EML4:ALK}	
EVT-P-0026657-T03-IM6-161	P-0026657-T03-IM6		ALK-EML4	EML4	unknown	three_prime	29446952	20	1058	True	retained	1.0	False	Protein kinase domain		EML4 NA (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-5 fused to ALK exons 20-29): c.2312:EML4_c.3173-558:ALKInv Protein Fusion: mid-exon {EML4:ALK}	

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_1	breakpoint_2	breakpoint_3	protein_1	protein_2	breakpoint_4	is_main_ref	is_disrupted	is_disrupted	description	annotation	effect_on_fra
EVT-P-0015152-T02-IM6-162	P-0015152-T02-IM6		ALK- EML4	EML4	in-frame	three_prime	29447079	20	1058	True	retained	1.0	False	Protein kinase domain		EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-20 fused in- frame to ALK ex- ons 20-29): c.2242+127- 685:ALKInv Protein fusion: in frame (EML4- ALK) EML4 (NM_019063) - ALK (NM_004304) Fu- sion(EML4 exon 11 fused to ALK exon20): c.1489+671- 837:ALKInv Protein Fusion: in frame {EML4:ALK} EML4 (NM_01145076) - ALK (NM_004304) fusion (EML4 exons 1- 12 fused with ALK exons 20-29): c.1316- 837:EML4_c.3173- 317:ALKInv Protein Fusion: in frame {EML4:ALK} EML4 (NM_019063) - ALK (NM_004304) fusion: c.1490- 2122:EML4_c.3172+564:ALKInv Protein Fusion: in frame {EML4:ALK} EML4 (NM_001145076) - ALK (NM_004304) fusion (EML4 exons 1-2 fused with ALK ex- ons 20-29): c.208+1245- 647:ALKInv Protein Fusion: in frame {EML4:ALK}	NA EML4_c.3173- NA EML4_c.3172+839:ALK NA EML4_c.3173- NA EML4_c.3172+564:ALKInv NA EML4_c.3172+647:ALKInv	
EVT-P-0032862-T01-IM6-163	P-0032862-T01-IM6		ALK- EML4	EML4	in-frame	three_prime	29447488	19	1058	True	retained	1.0	False	Protein kinase domain		EML4 (NM_019063) - ALK (NM_004304) Fu- sion(EML4 exon 11 fused to ALK exon20): c.1489+671- 837:ALKInv Protein Fusion: in frame {EML4:ALK} EML4 (NM_01145076) - ALK (NM_004304) fusion (EML4 exons 1- 12 fused with ALK exons 20-29): c.1316- 837:EML4_c.3173- 317:ALKInv Protein Fusion: in frame {EML4:ALK} EML4 (NM_019063) - ALK (NM_004304) fusion: c.1490- 2122:EML4_c.3172+564:ALKInv Protein Fusion: in frame {EML4:ALK} EML4 (NM_001145076) - ALK (NM_004304) fusion (EML4 exons 1-2 fused with ALK ex- ons 20-29): c.208+1245- 647:ALKInv Protein Fusion: in frame {EML4:ALK}	NA EML4_c.3172+839:ALK NA EML4_c.3173- NA EML4_c.3172+564:ALKInv NA EML4_c.3172+647:ALKInv	
EVT-P-0027119-T03-IM6-164	P-0027119-T03-IM6		ALK- EML4	EML4	in-frame	three_prime	29446711	20	1058	True	retained	1.0	False	Protein kinase domain		EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1- 12 fused with ALK exons 20-29): c.1316- 837:EML4_c.3173- 317:ALKInv Protein Fusion: in frame {EML4:ALK} EML4 (NM_01145076) - ALK (NM_004304) fusion (EML4 exons 1- 12 fused with ALK exons 20-29): c.1316- 837:EML4_c.3173- 317:ALKInv Protein Fusion: in frame {EML4:ALK} EML4 (NM_019063) - ALK (NM_004304) fusion: c.1490- 2122:EML4_c.3172+564:ALKInv Protein Fusion: in frame {EML4:ALK} EML4 (NM_001145076) - ALK (NM_004304) fusion (EML4 exons 1-2 fused with ALK ex- ons 20-29): c.208+1245- 647:ALKInv Protein Fusion: in frame {EML4:ALK}	NA EML4_c.3173- NA EML4_c.3172+564:ALKInv NA EML4_c.3172+647:ALKInv	
EVT-P-0032683-T01-IM6-165	P-0032683-T01-IM6		ALK- EML4	EML4	in-frame	three_prime	29447763	19	1058	True	retained	1.0	False	Protein kinase domain		EML4 (NM_019063) - ALK (NM_004304) fusion: c.1490- 2122:EML4_c.3172+564:ALKInv Protein Fusion: in frame {EML4:ALK} EML4 (NM_019063) - ALK (NM_004304) fusion: c.1490- 2122:EML4_c.3172+564:ALKInv Protein Fusion: in frame {EML4:ALK} EML4 (NM_001145076) - ALK (NM_004304) fusion (EML4 exons 1-2 fused with ALK ex- ons 20-29): c.208+1245- 647:ALKInv Protein Fusion: in frame {EML4:ALK}	NA EML4_c.3172+564:ALKInv NA EML4_c.3172+647:ALKInv	
EVT-P-0028339-T01-IM6-166	P-0028339-T01-IM6		ALK- EML4	EML4	in-frame	three_prime	29447680	19	1058	True	retained	1.0	False	Protein kinase domain		EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-2 fused with ALK ex- ons 20-29): c.208+1245- 647:ALKInv Protein Fusion: in frame {EML4:ALK}	NA EML4_c.3172+647:ALKInv	

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_1	breakpoint_2	breakpoint_3	protein_1	protein_2	breakpoint_4	is_main_ref	is_disrupted	is_lost	is_domain	disrupted_domain	description	annotation	effect_on_fra
EVT-P-0018541-T01-IM6-174	P-0018541-T01-IM6		ALK-LRP1B	LRP1B	in-frame	five_prime	29450289	17	972	True	lost	0.0	False			Protein kinase domain		ALK (NM_004304) - LRP1B (NM_018557) Rearrangement: c.2914+151:ALK_c.344-85984:LRP1Bdup Protein Fusion: in frame {ALK:LRP1B}	NA	
EVT-P-0027092-T01-IM6-176	P-0027092-T01-IM6		ALK-PLEKHH2	PLEKHH2	in-frame	three_prime	29446506	20	1058	True	retained	1.0	False			Protein kinase domain		PLEKHH2 (NM_172069) - ALK (NM_004304) rearrangement: c.502+545:PLEKHH2_c.3173-112:ALKinv Protein Fusion: in frame {ALK:LRP1B}	NA	
EVT-P-0027363-T02-IM6-177	P-0027363-T02-IM6		ALK-RANBP2	RANBP2	in-frame	three_prime	29447364	19	1058	True	retained	1.0	False			Protein kinase domain		RANBP2 (NM_006267) - ALK (NM_004304) rearrangement: c.2603-1556:RANBP2_c.3172+963:ALKinv Protein Fusion: in frame {PLEKHH2:ALK}	NA	
EVT-P-0068865-T02-IM7-178	P-0068865-T02-IM7		ALK-RANBP2	RANBP2	in-frame	five_prime	29447373	19	1058	True	lost	0.0	False			Protein kinase domain		ALK (NM_004304) - RANBP2 (NM_006267) rearrangement: c.3172+954:ALK_c.2602+1532:RANBP2 Protein Fusion: in frame {RANBP2:ALK}	NA	
EVT-P-0008843-T01-IM5-179	P-0008843-T01-IM5		ALK-RANBP2	RANBP2	in-frame	three_prime	29446488	20	1058	True	retained	1.0	False			Protein kinase domain		ALK (NM_004304) - RANBP2 (NM_006267) rearrangement: c.2603-1070:RANBP2_c.3173-94:ALKinv Protein fusion: in frame {ALK:RANBP2}	NA	
EVT-P-0015699-T01-IM6-181	P-0015699-T01-IM6		ALK-SPTBN1	SPTBN1	in-frame	three_prime	29447437	19	1058	True	retained	1.0	False			Protein kinase domain		SPTBN1 (NM_003128) - ALK (NM_004304) fusion (SPTBN1 exon 7 fused to ALK exon 20): c.764-1028:SPTBN1_c.3172+890:ALKinv Protein fusion: in frame (SPTBN1-ALK)	NA	

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_1	breakpoint_2	breakpoint_3	protein_1_pos	protein_2_pos	breakpoint_4	is_main_ref	is_in_frame	is_retained	is_in_domain	disrupted_domains	description_annotation	effect_on_fra
EVT-P-0016193-T05-IM6-182	P-0016193-T05-IM6		ALK-STRN	STRN	unknown	three_prime	29448326	19	1058	True	retained	1.0	False	Protein kinase domain				STRN (NM_003162) - ALK (NM_004304) fusion (STRN exons 1-3 fused with ALK exons 20-29): c.412+4919:STRN_c.3172+1:ALKdel	NA
EVT-P-0016193-T02-IM6-183	P-0016193-T02-IM6		ALK-STRN	STRN	unknown	three_prime	29448326	19	1058	True	retained	1.0	False	Protein kinase domain				STRN (NM_003162) - ALK (NM_004304) fusion (STRN exons 1-3 fused with ALK exons 20-29): c.412+4919:STRNdel	NA
EVT-P-0016193-T01-IM6-184	P-0016193-T01-IM6		ALK-STRN	STRN	unknown	three_prime	29448326	19	1058	True	retained	1.0	False	Protein kinase domain				STRN (NM_003162) - ALK (NM_004304) fusion (STRN exons 1-3 fused with ALK exons 20-29): c.412+4919:STRNdel	NA
EVT-P-0041438-T01-IM6-185	P-0041438-T01-IM6		ALK-TTC27	TTC27	out-of-frame	five_prime	29447817	19	1058	True	lost	0.0	False	Protein kinase domain				STRN (NM_003162) - TTC27 (NM_017735) rearrangement: c.3172+510:ALK_c.538-4165:TTC27inv	NA
EVT-P-0006617-T01-IM5-187	P-0006617-T01-IM5		ASAP2-ALK	ASAP2	unknown	five_prime	30142753	1	223	True	lost	0.0	False	Protein kinase domain				ASAP2 (NM_003887) - ALK (NM_004304) rearrangement event : c.127-20148:ASAP2_c.667+106:ALKdel	NA
EVT-P-0062994-T02-IM7-188	P-0062994-T02-IM7		CLTC-ALK	CLTC	in-frame	three_prime	29449624	18	1023	True	retained	1.0	False	Protein kinase domain				CLTC (NM_004859) - ALK (NM_004304) fusion: t(2;17)(p23;q23.1)(chr2:g.29449624::chr2	

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_1	breakpoint_2	breakpoint_3	protein_pos	protein_pos	breakpoint_4	main_retention	donor_inframe	truncated	domain	disrupted_domains	annotation	effect_on_fusion
EVT-P-0027119-T04-IM7-197	P-0027119-T04-IM7		EML4-ALK	EML4	in-frame	three_prime	29446711	20	1058	True	retained	1.0	False	Protein kinase domain				EML4 (NM_019063) - ALK (NM_004304) fusion: c.1490-837:EML4_c.3173-317:ALKInv Protein Fusion: in frame {EML4:ALK} EML4 (NM_019063) - ALK (NM_004304) fusion: c.668-3018:EML4_c.3173-698:ALKInv Protein Fusion: in frame {EML4:ALK} EML4 (NM_019063) - ALK (NM_004304) fusion: c.667+4909:EML4_c.3173-176:ALKInv Protein Fusion: in frame {EML4:ALK} EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-20 fused to ALK exons 20 -29): c.2243-278:EML4_c.3173-927:ALKInv Protein Fusion: in frame {EML4:ALK} EML4 (NM_001145076) - ALK (NM_004304) fusion (EML4 exon 12 fused to ALK exon 20): c.1316-1940:EML4_c.3173-758:ALKInv Protein Fusion: in frame {EML4:ALK} EML4 (NM_019063) - ALK (NM_004304) fusion: c.1594:EML4_c.3172+910:ALKInv Protein Fusion: mid-exon {EML4:ALK}	
EVT-P-0057391-T02-IM6-198	P-0057391-T02-IM6		EML4-ALK	EML4	in-frame	three_prime	29447092	20	1058	True	retained	1.0	False	Protein kinase domain				EML4 (NM_019063) - ALK (NM_004304) fusion: c.668-3018:EML4_c.3173-698:ALKInv Protein Fusion: in frame {EML4:ALK} EML4 (NM_019063) - ALK (NM_004304) fusion: c.667+4909:EML4_c.3173-176:ALKInv Protein Fusion: in frame {EML4:ALK} EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-20 fused to ALK exons 20 -29): c.2243-278:EML4_c.3173-927:ALKInv Protein Fusion: in frame {EML4:ALK} EML4 (NM_001145076) - ALK (NM_004304) fusion (EML4 exon 12 fused to ALK exon 20): c.1316-1940:EML4_c.3173-758:ALKInv Protein Fusion: in frame {EML4:ALK} EML4 (NM_019063) - ALK (NM_004304) fusion: c.1594:EML4_c.3172+910:ALKInv Protein Fusion: mid-exon {EML4:ALK}	
EVT-P-0034334-T01-IM6-199	P-0034334-T01-IM6		EML4-ALK	EML4	in-frame	three_prime	29446570	20	1058	True	retained	1.0	False	Protein kinase domain				EML4 (NM_019063) - ALK (NM_004304) fusion: c.668-3018:EML4_c.3173-698:ALKInv Protein Fusion: in frame {EML4:ALK} EML4 (NM_019063) - ALK (NM_004304) fusion: c.667+4909:EML4_c.3173-176:ALKInv Protein Fusion: in frame {EML4:ALK} EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-20 fused to ALK exons 20 -29): c.2243-278:EML4_c.3173-927:ALKInv Protein Fusion: in frame {EML4:ALK} EML4 (NM_001145076) - ALK (NM_004304) fusion (EML4 exon 12 fused to ALK exon 20): c.1316-1940:EML4_c.3173-758:ALKInv Protein Fusion: in frame {EML4:ALK} EML4 (NM_019063) - ALK (NM_004304) fusion: c.1594:EML4_c.3172+910:ALKInv Protein Fusion: mid-exon {EML4:ALK}	
EVT-P-0025095-T01-IM6-200	P-0025095-T01-IM6		EML4-ALK	EML4	in-frame	three_prime	29447321	20	1058	True	retained	1.0	False	Protein kinase domain				EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-20 fused to ALK exons 20 -29): c.2243-278:EML4_c.3173-927:ALKInv Protein Fusion: in frame {EML4:ALK} EML4 (NM_019063) - ALK (NM_004304) fusion: c.668-3018:EML4_c.3173-698:ALKInv Protein Fusion: in frame {EML4:ALK} EML4 (NM_019063) - ALK (NM_004304) fusion: c.667+4909:EML4_c.3173-176:ALKInv Protein Fusion: in frame {EML4:ALK} EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-20 fused to ALK exons 20 -29): c.2243-278:EML4_c.3173-927:ALKInv Protein Fusion: in frame {EML4:ALK} EML4 (NM_001145076) - ALK (NM_004304) fusion (EML4 exon 12 fused to ALK exon 20): c.1316-1940:EML4_c.3173-758:ALKInv Protein Fusion: in frame {EML4:ALK} EML4 (NM_019063) - ALK (NM_004304) fusion: c.1594:EML4_c.3172+910:ALKInv Protein Fusion: mid-exon {EML4:ALK}	
EVT-P-0025040-T01-IM6-201	P-0025040-T01-IM6		EML4-ALK	EML4	in-frame	three_prime	29447152	20	1058	True	retained	1.0	False	Protein kinase domain				EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exon 12 fused to ALK exon 20): c.1316-1940:EML4_c.3173-758:ALKInv Protein Fusion: in frame {EML4:ALK} EML4 (NM_019063) - ALK (NM_004304) fusion: c.668-3018:EML4_c.3173-698:ALKInv Protein Fusion: in frame {EML4:ALK} EML4 (NM_019063) - ALK (NM_004304) fusion: c.667+4909:EML4_c.3173-176:ALKInv Protein Fusion: in frame {EML4:ALK} EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-20 fused to ALK exons 20 -29): c.2243-278:EML4_c.3173-927:ALKInv Protein Fusion: in frame {EML4:ALK} EML4 (NM_001145076) - ALK (NM_004304) fusion (EML4 exon 12 fused to ALK exon 20): c.1316-1940:EML4_c.3173-758:ALKInv Protein Fusion: in frame {EML4:ALK} EML4 (NM_019063) - ALK (NM_004304) fusion: c.1594:EML4_c.3172+910:ALKInv Protein Fusion: mid-exon {EML4:ALK}	
EVT-P-0007705-T02-IM6-202	P-0007705-T02-IM6		EML4-ALK	EML4	unknown	three_prime	29447417	19	1058	True	retained	1.0	False	Protein kinase domain				EML4 (NM_019063) - ALK (NM_004304) fusion: c.1594:EML4_c.3172+910:ALKInv Protein Fusion: mid-exon {EML4:ALK}	

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_1	breakpoint_2	breakpoint_3	protein_ratio	breakpoint_4	main_ref	donor_frame	main_ref	disrupted_domain	disrupted_domain	description	annotation	effect_on_fra
EVT-P-0040574-T01-IM6-223	P-0040574-T01-IM6		EML4-ALK	EML4	in-frame	three_prime	29447738	19	1058	True	retained	1.0	False	Protein kinase domain			EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exon5 fused with ALK exon20) : c.668-1589:EML4_c.3172+589:ALKInv Protein Fusion: in frame {EML4:ALK}	NA	
EVT-P-0040576-T01-IM6-224	P-0040576-T01-IM6		EML4-ALK	EML4	in-frame	three_prime	29447321	20	1058	True	retained	1.0	False	Protein kinase domain			EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exon5 fused with ALK exon20) : c.668-821:EML4_c.3173-927:ALKInv Protein Fusion: in frame {EML4:ALK}	NA	
EVT-P-0040600-T01-IM6-225	P-0040600-T01-IM6		EML4-ALK	EML4	in-frame	three_prime	29447688	19	1058	True	retained	1.0	False	Protein kinase domain			EML4 (NM_019063) - ALK (NM_004304) fusion: c.667+1795:EML4_c.3172+639:ALKInv Protein Fusion: in frame {EML4:ALK}	NA	
EVT-P-0032683-T02-IM7-226	P-0032683-T02-IM7		EML4-ALK	EML4	in-frame	three_prime	29447763	19	1058	True	retained	1.0	False	Protein kinase domain			EML4 (NM_019063) - ALK (NM_004304) fusion: c.1490-2122:EML4_c.3172+564:ALKInv Protein Fusion: in frame {EML4:ALK}	NA	
EVT-P-0014385-T06-IM7-227	P-0014385-T06-IM7		EML4-ALK	EML4	in-frame	three_prime	29447676	19	1058	True	retained	1.0	False	Protein kinase domain			EML4 (NM_019063) - ALK (NM_004304) fusion: c.1489+1890:EML4_c.3172+651:ALKInv Protein Fusion: in frame {EML4:ALK}	NA	
EVT-P-0067069-T01-IM7-228	P-0067069-T01-IM7		EML4-ALK	EML4	in-frame	three_prime	29448169	19	1058	True	retained	1.0	False	Protein kinase domain			EML4 (NM_019063) - ALK (NM_004304) fusion: c.1489+2305:EML4_c.3172+158:ALKInv Protein Fusion: in frame {EML4:ALK}	NA	
EVT-P-0014385-T02-IM6-229	P-0014385-T02-IM6		EML4-ALK	EML4	in-frame	three_prime	29447676	19	1058	True	retained	1.0	False	Protein kinase domain			EML4 (NM_019063) - ALK (NM_004304) fusion: c.1489+1890:EML4_c.3172+651:ALKInv Protein Fusion: in frame {EML4:ALK}	NA	

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_1	breakpoint_2	breakpoint_3	protein_1	protein_2	breakpoint_4	main_ref	is_in_frame	is_inverted	is_in_domain	disrupted_domain	description	annotation	effect_on_fra
EVT-P-0012913-T02-IM6-237	P-0012913-T02-IM6		EML4-ALK	EML4	in-frame	three_prime	29446965	20	1058	True	retained	1.0	False	Protein kinase domain				EML4 (NM_019063) - ALK (NM_004304) fusion: c.209-4133:EML4_c.3173-571:ALKInv Protein Fusion: in frame {EML4:ALK}	NA	
EVT-P-0043350-T01-IM6-238	P-0043350-T01-IM6		EML4-ALK	EML4	in-frame	three_prime	29447513	19	1058	True	retained	1.0	False	Protein kinase domain				EML4 (NM_019063) - ALK (NM_004304) fusion: c.668-4017:EML4_c.3172+814:ALKInv Protein Fusion: in frame {EML4:ALK}	NA	
EVT-P-0043363-T01-IM6-239	P-0043363-T01-IM6		EML4-ALK	EML4	unknown	three_prime	29448101	19	1058	True	retained	1.0	False	Protein kinase domain				EML4 (NM_019063) - ALK (NM_004304) fusion: c.2254:EML4_c.3172+226:ALKInv Protein Fusion: mid-exon {EML4:ALK}	NA	
EVT-P-0043594-T01-IM6-240	P-0043594-T01-IM6		EML4-ALK	EML4	unknown	three_prime	29448381	19	1040	False	retained	1.0	False	Protein kinase domain				EML4 (NM_019063) - ALK (NM_004304) fusion: c.339-2119:EML4_c.3118:ALKInv Protein Fusion: mid-exon {EML4:ALK}	NA	
EVT-P-0069396-T02-IM7-241	P-0069396-T02-IM7		EML4-ALK	EML4	in-frame	three_prime	29446944	20	1058	True	retained	1.0	False	Protein kinase domain				EML4 (NM_019063) - ALK (NM_004304) fusion: c.668-4308:EML4_c.3173-550:ALKInv Protein Fusion: in frame {EML4:ALK}	NA	
EVT-P-0069535-T01-IM7-242	P-0069535-T01-IM7		EML4-ALK	EML4	in-frame	three_prime	29447764	19	1058	True	retained	1.0	False	Protein kinase domain				EML4 (NM_019063) - ALK (NM_004304) fusion: c.208+3829:EML4_c.3172+563:ALKInv Protein Fusion: in frame {EML4:ALK}	NA	
EVT-P-0069617-T01-IM7-243	P-0069617-T01-IM7		EML4-ALK	EML4	in-frame	three_prime	29448017	19	1058	True	retained	1.0	False	Protein kinase domain				EML4 (NM_019063) - ALK (NM_004304) fusion: c.667+7821:EML4_c.3172+310:ALKInv Protein Fusion: in frame {EML4:ALK}	NA	

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_1	breakpoint_2	breakpoint_3	protein_pos	breakpoint_4	domain_retained	domain_lost	domain_deleted	disrupted_domains	sequence_annotation	clinical_effect_on_patient
EVT-P-0044432-T01-IM6-244	P-0044432-T01-IM6		EML4-ALK	EML4	unknown	three_prime	29448344	19	1052	False	retained	1.0	False	Protein kinase domain		EML4 (NM_019063) - ALK (NM_004304) fusion: c.2057-576:EML4_c.3155:ALKInv Protein Fusion: mid-exon {EML4:ALK} EML4 NA (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-20 with ALK exons 20-29) : c.2243-181:EML4_c.3172+658:ALKInv Protein fusion: in frame (EML4-ALK) EML4 NA (NM_019063) - ALK (NM_004304) fusion: c.2277:EML4_c.3172+744:ALKInv Protein Fusion: mid-exon {EML4:ALK} EML4 NA (NM_019063) - ALK (NM_004304) fusion: c.668-3064:EML4_c.3172+29:ALKInv Protein Fusion: in frame {EML4:ALK} EML4 NA (NM_019063) - ALK (NM_004304) fusion: c.668-4434:EML4_c.3172+407:ALKInv Protein Fusion: in frame {EML4:ALK} EML4 NA (NM_019063) - ALK (NM_004304) fusion: c.668-1949:EML4_c.3173-171:ALKInv Protein Fusion: in frame {EML4:ALK}	
EVT-P-0005743-T02-IM5-245	P-0005743-T02-IM5		EML4-ALK	EML4	in-frame	three_prime	29447669	19	1058	True	retained	1.0	False	Protein kinase domain		EML4 NA (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-20 with ALK exons 20-29) : c.2243-181:EML4_c.3172+658:ALKInv Protein fusion: in frame (EML4-ALK) EML4 NA (NM_019063) - ALK (NM_004304) fusion: c.2277:EML4_c.3172+744:ALKInv Protein Fusion: mid-exon {EML4:ALK} EML4 NA (NM_019063) - ALK (NM_004304) fusion: c.668-3064:EML4_c.3172+29:ALKInv Protein Fusion: in frame {EML4:ALK} EML4 NA (NM_019063) - ALK (NM_004304) fusion: c.668-4434:EML4_c.3172+407:ALKInv Protein Fusion: in frame {EML4:ALK}	
EVT-P-0045222-T03-IM7-246	P-0045222-T03-IM7		EML4-ALK	EML4	unknown	three_prime	29447583	19	1058	True	retained	1.0	False	Protein kinase domain		EML4 NA (NM_019063) - ALK (NM_004304) fusion: c.2277:EML4_c.3172+744:ALKInv Protein Fusion: mid-exon {EML4:ALK} EML4 NA (NM_019063) - ALK (NM_004304) fusion: c.668-3064:EML4_c.3172+29:ALKInv Protein Fusion: in frame {EML4:ALK} EML4 NA (NM_019063) - ALK (NM_004304) fusion: c.668-4434:EML4_c.3172+407:ALKInv Protein Fusion: in frame {EML4:ALK}	
EVT-P-0045243-T01-IM6-247	P-0045243-T01-IM6		EML4-ALK	EML4	in-frame	three_prime	29448298	19	1058	True	retained	1.0	False	Protein kinase domain		EML4 NA (NM_019063) - ALK (NM_004304) fusion: c.668-3064:EML4_c.3172+29:ALKInv Protein Fusion: in frame {EML4:ALK} EML4 NA (NM_019063) - ALK (NM_004304) fusion: c.668-4434:EML4_c.3172+407:ALKInv Protein Fusion: in frame {EML4:ALK}	
EVT-P-0045341-T01-IM6-248	P-0045341-T01-IM6		EML4-ALK	EML4	in-frame	three_prime	29447920	19	1058	True	retained	1.0	False	Protein kinase domain		EML4 NA (NM_019063) - ALK (NM_004304) fusion: c.668-4434:EML4_c.3172+407:ALKInv Protein Fusion: in frame {EML4:ALK} EML4 NA (NM_019063) - ALK (NM_004304) fusion: c.668-1949:EML4_c.3173-171:ALKInv Protein Fusion: in frame {EML4:ALK}	
EVT-P-0065469-T02-IM7-249	P-0065469-T02-IM7		EML4-ALK	EML4	in-frame	three_prime	29446565	20	1058	True	retained	1.0	False	Protein kinase domain		EML4 NA (NM_019063) - ALK (NM_004304) fusion: c.668-1949:EML4_c.3173-171:ALKInv Protein Fusion: in frame {EML4:ALK}	
EVT-P-0065210-T01-IM7-250	P-0065210-T01-IM7		EML4-ALK	EML4	in-frame	three_prime	29448076	19	1058	True	retained	1.0	False	Protein kinase domain		EML4 NA (NM_019063) - ALK (NM_004304) fusion: c.668-2012:EML4_c.3172+251:ALKInv Protein Fusion: in frame {EML4:ALK}	

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_1	breakpoint_2	breakpoint_3	protein_1	protein_2	breakpoint_4	is_main_ref	is_disrupted	is_disrupted	description	annotation	effect_on_fusion
EVT-P-0012873-T03-IM7-251	P-0012873-T03-IM7		EML4-ALK	EML4	in-frame	three_prime	29446439	20	1058	True	retained	1.0	False	Protein kinase domain		EML4 (NM_019063) - ALK (NM_004304) fusion: c.1489+2231:45:ALKInv Protein	NA	
EVT-P-0064716-T02-IM7-252	P-0064716-T02-IM7		EML4-ALK	EML4	in-frame	three_prime	29447739	19	1058	True	retained	1.0	False	Protein kinase domain		Fusion: in frame {EML4:ALK} EML4 (NM_019063) - ALK (NM_004304) fusion: c.668-3669: EML4 Protein	NA	
EVT-P-0031368-T04-IM6-253	P-0031368-T04-IM6		EML4-ALK	EML4	in-frame	three_prime	29447191	20	1058	True	retained	1.0	False	Protein kinase domain		Fusion: in frame {EML4:ALK} EML4 (NM_019063) - ALK (NM_004304) fusion: c.1490-1154: EML4 Protein	NA	
EVT-P-0045912-T02-IM6-254	P-0045912-T02-IM6		EML4-ALK	EML4	out-of-frame	three_prime	29446507	20	1058	True	retained	1.0	False	Protein kinase domain		Fusion: in frame {EML4:ALK} EML4 (NM_019063) - ALK (NM_004304) fusion: c.2154+1011: EML4 Protein	NA	
EVT-P-0031824-T01-IM6-255	P-0031824-T01-IM6		EML4-ALK	EML4	in-frame	three_prime	29450557	17	939	True	retained	1.0	False	Protein kinase domain		Fusion: out of frame {EML4:ALK} EML4 (NM_019063) - ALK (NM_004304) fusion: c.208+2971: EML4 Protein	NA	
EVT-P-0046083-T02-IM7-256	P-0046083-T02-IM7		EML4-ALK	EML4	in-frame	three_prime	29446665	20	1058	True	retained	1.0	False	Protein kinase domain		Fusion: in frame {EML4:ALK} EML4 (NM_019063) - ALK (NM_004304) fusion: c.1489+986: EML4 Protein	NA	
EVT-P-0046594-T01-IM6-257	P-0046594-T01-IM6		EML4-ALK	EML4	in-frame	three_prime	29446445	20	1058	True	retained	1.0	False	Protein kinase domain		Fusion: in frame {EML4:ALK} EML4 (NM_019063) - ALK (NM_004304) fusion: c.667+2217: EML4 Protein	NA	

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_1	breakpoint_2	breakpoint_3	protein_pos	protein_neg	breakpoint_4	main_retained	donor_fractions	truncated	protein_domains	disrupted_domains	describes_annotation	station_site	effect_on_fractions
EVT-P-0040732-T02-IM6-258	P-0040732-T02-IM6		EML4-ALK	EML4	in-frame	three_prime	29446636	20	1058	True	retained	1.0	False	Protein kinase domain				EML4 (NM_019063) - ALK (NM_004304) fusion: c.1490-1832:EML4_c.3173-242:ALKInv Protein Fusion: in frame {EML4:ALK} EML4 (NM_019063) - ALK (NM_004304) fusion: c.667+925:EML4_c.3173-215:ALKInv Protein Fusion: in frame {EML4:ALK} EML4 (NM_019063) - ALK (NM_004304) fusion: c.667+925:EML4_c.3173-215:ALKInv Protein Fusion: in frame {EML4:ALK} EML4 (NM_019063) - ALK (NM_004304) fusion: c.208+4042:EML4_c.3173-140:ALKInv Protein Fusion: in frame {EML4:ALK} EML4 (NM_019063) - ALK (NM_004304) fusion: c.668-2813:EML4_c.3173-490:ALKInv Protein Fusion: in frame {EML4:ALK} EML4 (NM_019063) - ALK (NM_004304) fusion: c.1489+2231:EML4_c.3173-45:ALKInv Protein Fusion: in frame {EML4:ALK} EML4 (NM_019063) - ALK (NM_004304) fusion: c.1490-2055:EML4_c.3173-697:ALKInv Protein Fusion: in frame {EML4:ALK}	NA	
EVT-P-0047355-T01-IM6-259	P-0047355-T01-IM6		EML4-ALK	EML4	in-frame	three_prime	29446609	20	1058	True	retained	1.0	False	Protein kinase domain				EML4 (NM_019063) - ALK (NM_004304) fusion: c.667+925:EML4_c.3173-215:ALKInv Protein Fusion: in frame {EML4:ALK} EML4 (NM_019063) - ALK (NM_004304) fusion: c.667+925:EML4_c.3173-215:ALKInv Protein Fusion: in frame {EML4:ALK} EML4 (NM_019063) - ALK (NM_004304) fusion: c.208+4042:EML4_c.3173-140:ALKInv Protein Fusion: in frame {EML4:ALK} EML4 (NM_019063) - ALK (NM_004304) fusion: c.668-2813:EML4_c.3173-490:ALKInv Protein Fusion: in frame {EML4:ALK} EML4 (NM_019063) - ALK (NM_004304) fusion: c.1489+2231:EML4_c.3173-45:ALKInv Protein Fusion: in frame {EML4:ALK} EML4 (NM_019063) - ALK (NM_004304) fusion: c.1490-2055:EML4_c.3173-697:ALKInv Protein Fusion: in frame {EML4:ALK}	NA	
EVT-P-0047355-T02-IM6-260	P-0047355-T02-IM6		EML4-ALK	EML4	in-frame	three_prime	29446609	20	1058	True	retained	1.0	False	Protein kinase domain				EML4 (NM_019063) - ALK (NM_004304) fusion: c.667+925:EML4_c.3173-215:ALKInv Protein Fusion: in frame {EML4:ALK} EML4 (NM_019063) - ALK (NM_004304) fusion: c.667+925:EML4_c.3173-215:ALKInv Protein Fusion: in frame {EML4:ALK} EML4 (NM_019063) - ALK (NM_004304) fusion: c.208+4042:EML4_c.3173-140:ALKInv Protein Fusion: in frame {EML4:ALK} EML4 (NM_019063) - ALK (NM_004304) fusion: c.668-2813:EML4_c.3173-490:ALKInv Protein Fusion: in frame {EML4:ALK} EML4 (NM_019063) - ALK (NM_004304) fusion: c.1489+2231:EML4_c.3173-45:ALKInv Protein Fusion: in frame {EML4:ALK} EML4 (NM_019063) - ALK (NM_004304) fusion: c.1490-2055:EML4_c.3173-697:ALKInv Protein Fusion: in frame {EML4:ALK}	NA	
EVT-P-0064342-T01-IM7-261	P-0064342-T01-IM7		EML4-ALK	EML4	in-frame	three_prime	29446534	20	1058	True	retained	1.0	False	Protein kinase domain				EML4 (NM_019063) - ALK (NM_004304) fusion: c.208+4042:EML4_c.3173-140:ALKInv Protein Fusion: in frame {EML4:ALK} EML4 (NM_019063) - ALK (NM_004304) fusion: c.668-2813:EML4_c.3173-490:ALKInv Protein Fusion: in frame {EML4:ALK} EML4 (NM_019063) - ALK (NM_004304) fusion: c.1489+2231:EML4_c.3173-45:ALKInv Protein Fusion: in frame {EML4:ALK} EML4 (NM_019063) - ALK (NM_004304) fusion: c.1490-2055:EML4_c.3173-697:ALKInv Protein Fusion: in frame {EML4:ALK}	NA	
EVT-P-0063585-T01-IM7-262	P-0063585-T01-IM7		EML4-ALK	EML4	in-frame	three_prime	29446884	20	1058	True	retained	1.0	False	Protein kinase domain				EML4 (NM_019063) - ALK (NM_004304) fusion: c.668-2813:EML4_c.3173-490:ALKInv Protein Fusion: in frame {EML4:ALK} EML4 (NM_019063) - ALK (NM_004304) fusion: c.1489+2231:EML4_c.3173-45:ALKInv Protein Fusion: in frame {EML4:ALK} EML4 (NM_019063) - ALK (NM_004304) fusion: c.1490-2055:EML4_c.3173-697:ALKInv Protein Fusion: in frame {EML4:ALK}	NA	
EVT-P-0012873-T02-IM6-263	P-0012873-T02-IM6		EML4-ALK	EML4	in-frame	three_prime	29446439	20	1058	True	retained	1.0	False	Protein kinase domain				EML4 (NM_019063) - ALK (NM_004304) fusion: c.1489+2231:EML4_c.3173-45:ALKInv Protein Fusion: in frame {EML4:ALK} EML4 (NM_019063) - ALK (NM_004304) fusion: c.1490-2055:EML4_c.3173-697:ALKInv Protein Fusion: in frame {EML4:ALK}	NA	
EVT-P-0062999-T02-IM7-264	P-0062999-T02-IM7		EML4-ALK	EML4	in-frame	three_prime	29447091	20	1058	True	retained	1.0	False	Protein kinase domain				EML4 (NM_019063) - ALK (NM_004304) fusion: c.1490-2055:EML4_c.3173-697:ALKInv Protein Fusion: in frame {EML4:ALK}	NA	

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_1	breakpoint_2	breakpoint_3	protein_1	protein_2	breakpoint_4	main_ref	donor_frame	inframe	disrupted_domain	disrupted_domain	description	annotation	effect_on_fra
EVT-P-0049318-T01-IM6-265	P-0049318-T01-IM6		EML4-ALK	EML4	unknown	three_prime	29446359	20	1070	False	retained	1.0	False	Protein kinase domain				EML4 (NM_019063) - ALK (NM_004304) fusion: c.2154+2335:EML4_c.3208:ALKInv Protein Fusion: mid-exon {EML4:ALK} EML4 (NM_019063) - ALK (NM_004304) fusion: c.1641+1:EML4_c.3173-275:ALKInv Protein Fusion: out of frame {EML4:ALK} EML4 (NM_019063) - ALK (NM_004304) fusion: c.667+1166:EML4_c.3173-338:ALKInv Protein Fusion: in frame {EML4:ALK} EML4 (NM_019063) - ALK (NM_004304) fusion: c.2242+9:EML4_c.3173-597:ALKInv Protein Fusion: in frame {EML4:ALK} EML4 (NM_019063) - ALK (NM_004304) fusion: c.1490-2649:EML4_c.3173-964:ALKInv Protein Fusion: in frame {EML4:ALK} EML4 (NM_019063) - ALK (NM_004304) fusion: c.1490-639:EML4_c.3173-252:ALKInv Protein Fusion: in frame {EML4:ALK} EML4 (NM_019063) - ALK (NM_004304) fusion: c.667+4130:EML4_c.3173-380:ALKInv Protein Fusion: in frame {EML4:ALK}	NA	
EVT-P-0046720-T01-IM6-266	P-0046720-T01-IM6		EML4-ALK	EML4	out-of-frame	three_prime	29446669	20	1058	True	retained	1.0	False	Protein kinase domain				EML4 (NM_019063) - ALK (NM_004304) fusion: c.1641+1:EML4_c.3173-275:ALKInv Protein Fusion: out of frame {EML4:ALK} EML4 (NM_019063) - ALK (NM_004304) fusion: c.667+1166:EML4_c.3173-338:ALKInv Protein Fusion: in frame {EML4:ALK} EML4 (NM_019063) - ALK (NM_004304) fusion: c.2242+9:EML4_c.3173-597:ALKInv Protein Fusion: in frame {EML4:ALK} EML4 (NM_019063) - ALK (NM_004304) fusion: c.1490-2649:EML4_c.3173-964:ALKInv Protein Fusion: in frame {EML4:ALK} EML4 (NM_019063) - ALK (NM_004304) fusion: c.1490-639:EML4_c.3173-252:ALKInv Protein Fusion: in frame {EML4:ALK} EML4 (NM_019063) - ALK (NM_004304) fusion: c.667+4130:EML4_c.3173-380:ALKInv Protein Fusion: in frame {EML4:ALK}	NA	
EVT-P-0049375-T02-IM6-267	P-0049375-T02-IM6		EML4-ALK	EML4	in-frame	three_prime	29446732	20	1058	True	retained	1.0	False	Protein kinase domain				EML4 (NM_019063) - ALK (NM_004304) fusion: c.667+1166:EML4_c.3173-338:ALKInv Protein Fusion: in frame {EML4:ALK} EML4 (NM_019063) - ALK (NM_004304) fusion: c.2242+9:EML4_c.3173-597:ALKInv Protein Fusion: in frame {EML4:ALK} EML4 (NM_019063) - ALK (NM_004304) fusion: c.1490-2649:EML4_c.3173-964:ALKInv Protein Fusion: in frame {EML4:ALK} EML4 (NM_019063) - ALK (NM_004304) fusion: c.1490-639:EML4_c.3173-252:ALKInv Protein Fusion: in frame {EML4:ALK} EML4 (NM_019063) - ALK (NM_004304) fusion: c.667+4130:EML4_c.3173-380:ALKInv Protein Fusion: in frame {EML4:ALK}	NA	
EVT-P-0049511-T01-IM6-268	P-0049511-T01-IM6		EML4-ALK	EML4	in-frame	three_prime	29446991	20	1058	True	retained	1.0	False	Protein kinase domain				EML4 (NM_019063) - ALK (NM_004304) fusion: c.2242+9:EML4_c.3173-597:ALKInv Protein Fusion: in frame {EML4:ALK} EML4 (NM_019063) - ALK (NM_004304) fusion: c.1490-2649:EML4_c.3173-964:ALKInv Protein Fusion: in frame {EML4:ALK} EML4 (NM_019063) - ALK (NM_004304) fusion: c.1490-639:EML4_c.3173-252:ALKInv Protein Fusion: in frame {EML4:ALK} EML4 (NM_019063) - ALK (NM_004304) fusion: c.667+4130:EML4_c.3173-380:ALKInv Protein Fusion: in frame {EML4:ALK}	NA	
EVT-P-0010177-T02-IM7-269	P-0010177-T02-IM7		EML4-ALK	EML4	in-frame	three_prime	29447358	20	1058	True	retained	1.0	False	Protein kinase domain				EML4 (NM_019063) - ALK (NM_004304) fusion: c.1490-2649:EML4_c.3173-964:ALKInv Protein Fusion: in frame {EML4:ALK} EML4 (NM_019063) - ALK (NM_004304) fusion: c.1490-639:EML4_c.3173-252:ALKInv Protein Fusion: in frame {EML4:ALK} EML4 (NM_019063) - ALK (NM_004304) fusion: c.667+4130:EML4_c.3173-380:ALKInv Protein Fusion: in frame {EML4:ALK}	NA	
EVT-P-0031827-T03-IM6-270	P-0031827-T03-IM6		EML4-ALK	EML4	in-frame	three_prime	29446646	20	1058	True	retained	1.0	False	Protein kinase domain				EML4 (NM_019063) - ALK (NM_004304) fusion: c.1490-639:EML4_c.3173-252:ALKInv Protein Fusion: in frame {EML4:ALK} EML4 (NM_019063) - ALK (NM_004304) fusion: c.667+4130:EML4_c.3173-380:ALKInv Protein Fusion: in frame {EML4:ALK}	NA	
EVT-P-0051231-T01-IM6-271	P-0051231-T01-IM6		EML4-ALK	EML4	in-frame	three_prime	29446774	20	1058	True	retained	1.0	False	Protein kinase domain				EML4 (NM_019063) - ALK (NM_004304) fusion: c.667+4130:EML4_c.3173-380:ALKInv Protein Fusion: in frame {EML4:ALK}	NA	

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint	breakpoint	breakpoint	protein	protein	breakpoint	main_ref	chr	start	end	domain	disrupted_domain	description	annotation	effect_on_fra
EVT-P-0060148-T01-IM7-279	P-0060148-T01-IM7		EML4-ALK	EML4	in-frame	three_prime	29447757	19	1058	True	retained	1.0	False	Protein kinase domain					EML4 (NM_019063) - ALK (NM_004304) fusion: c.1489+2600: EML4_c.3172+570: ALKInv	NA	
EVT-P-0059699-T03-IM7-280	P-0059699-T03-IM7		EML4-ALK	EML4	in-frame	three_prime	29447467	19	1058	True	retained	1.0	False	Protein kinase domain					EML4 (NM_019063) - ALK (NM_004304) fusion: c.1490-1990: EML4_c.3172+860: ALKInv	NA	
EVT-P-0031827-T06-IM7-281	P-0031827-T06-IM7		EML4-ALK	EML4	in-frame	three_prime	29446646	20	1058	True	retained	1.0	False	Protein kinase domain					EML4 (NM_019063) - ALK (NM_004304) fusion: c.1490-639: EML4_c.3173-252: ALKInv	NA	
EVT-P-0054263-T01-IM6-282	P-0054263-T01-IM6		EML4-ALK	EML4	in-frame	three_prime	29446557	20	1058	True	retained	1.0	False	Protein kinase domain					EML4 (NM_019063) - ALK (NM_004304) fusion: c.668-210: EML4_c.3173-163: ALKInv	NA	
EVT-P-0059257-T02-IM7-283	P-0059257-T02-IM7		EML4-ALK	EML4	in-frame	three_prime	29448210	19	1058	True	retained	1.0	False	Protein kinase domain					EML4 (NM_019063) - ALK (NM_004304) fusion: c.668-982: EML4_c.3172+117: ALKInv	NA	
EVT-P-0054408-T01-IM6-284	P-0054408-T01-IM6		EML4-ALK	EML4	in-frame	three_prime	29446902	20	1058	True	retained	1.0	False	Protein kinase domain					EML4 (NM_019063) - ALK (NM_004304) fusion: c.2243-262: EML4_c.3173-508: ALKInv	NA	
EVT-P-0008332-T02-IM6-285	P-0008332-T02-IM6		EML4-ALK	EML4	unknown	three_prime	29446913	20	1058	True	retained	1.0	False	Protein kinase domain					EML4 (NM_001145076) - ALK (NM_004304) fusion: c.1315+828: EML4_c.3173-519: ALKdup	NA	

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_1	breakpoint_2	breakpoint_3	protein_1	protein_2	breakpoint_4	main_ref	chr1_start	chr1_end	chr2_start	chr2_end	disrupted_domain	description	annotation	effect_on_fra
EVT-P-0005711-T02-IM6-292	P-0005711-T02-IM6		KIF5B-ALK	KIF5B	in-frame	three_prime	29449548	18	1023	True	retained	1.0	False	Protein kinase domain					KIF5B (NM_004521) - ALK (NM_004304) fusion (KIF5B exons 1-24 fused with ALK exons 19-29) : t(2;10)(p23.2;p11.22)(chr2:g.29449548:::chr1:g.111761717::)	NA	
EVT-P-0008764-T01-IM5-293	P-0008764-T01-IM5		KIF5B-ALK	KIF5B	unknown	three_prime	29448327	19	1058	False	retained	1.0	False	Protein kinase domain					KIF5B (NM_004521) - ALK (NM_004304) fusion (KIF5B exons 1-24 fused with ALK exons 19-29) : t(2;10)(p23.2;p11.22)(chr2:g.29448327:::chr1:g.111761717::)	NA	
EVT-P-0017924-T01-IM6-294	P-0017924-T01-IM6		KIF5B-ALK	KIF5B	in-frame	three_prime	29447945	19	1058	True	retained	1.0	False	Protein kinase domain					KIF5B (NM_004521) - ALK (NM_004304) fusion : t(2;10)(p23.2;p11.22)(chr2:g.29447945:::chr1:g.111761717::)	NA	
EVT-P-0008764-T02-IM6-295	P-0008764-T02-IM6		KIF5B-ALK	KIF5B	unknown	three_prime	29448327	19	1058	False	retained	1.0	False	Protein kinase domain					Fusion {KIF5B:ALK} (KIF5B (NM_004521) - ALK (NM_004304) Fusion (KIF5B exons 1-24 fused with ALK exons 19-29) : t(2;10)(p23.2;p11.22)(chr2:g.29448327:::chr1:g.111761717::)	NA	
EVT-P-0067512-T01-IM7-297	P-0067512-T01-IM7		NOTCH1-ALK	NOTCH1	in-frame	three_prime	29449152	18	1023	True	retained	1.0	False	Protein kinase domain					NOTCH1 (NM_017617) - ALK (NM_004304) fusion : t(2;9)(p23.2;q34.3)(chr2:g.29449152:::chr1:g.111761717::)	NA	
EVT-P-0034044-T01-IM6-298	P-0034044-T01-IM6		NRP2-ALK	NRP2	in-frame	three_prime	29448550	19	1023	True	retained	1.0	False	Protein kinase domain					in frame {NOTCH1:ALK} NRP2 (NM_201266) - ALK (NM_004304) rearrangement: c.1291+575_c.3068-119inv Protein Fusion: in frame {NRP2:ALK}	NA	

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint	breakpoint	breakpoint	protein	protein	breakpoint	breakpoint	main_ref	domain	domain	domain	disrupted_domain	description	annotation	effect_on_fra
EVT-P-0055667-T06-IM7-299	P-0055667-T06-IM7		PICALM-ALK	PICALM	unknown	three_prime	29446386	20	1061	False	retained	1.0	False	Protein kinase domain					PICALM (NM_007166) - ALK (NM_004304) fusion: t(2;11)(p23;q14.2)(chr2:g.29446386::chr11:g.106100000::) Protein Fusion: mid-exon {PICALM:ALK} PICALM (NM_007166) - ALK (NM_004304) rearrangement: t(2;11)(p23;q14.2)(chr2:g.29446386::chr11:g.106100000::) Protein Fusion: mid-exon {PICALM:ALK}	NA	
EVT-P-0055667-T02-IM6-300	P-0055667-T02-IM6		PICALM-ALK	PICALM	unknown	three_prime	29446386	20	1061	False	retained	1.0	False	Protein kinase domain					PICALM (NM_007166) - ALK (NM_004304) fusion: t(2;11)(p23;q14.2)(chr2:g.29446386::chr11:g.106100000::) Protein Fusion: mid-exon {PICALM:ALK} PICALM (NM_007166) - ALK (NM_004304) rearrangement: t(2;11)(p23;q14.2)(chr2:g.29446386::chr11:g.106100000::) Protein Fusion: mid-exon {PICALM:ALK}	NA	
EVT-P-0055667-T04-IM6-301	P-0055667-T04-IM6		PICALM-ALK	PICALM	unknown	three_prime	29446386	20	1061	False	retained	1.0	False	Protein kinase domain					PICALM (NM_007166) - ALK (NM_004304) fusion: t(2;11)(p23;q14.2)(chr2:g.29446386::chr11:g.106100000::) Protein Fusion: mid-exon {PICALM:ALK} PICALM (NM_007166) - ALK (NM_004304) rearrangement: t(2;11)(p23;q14.2)(chr2:g.29446386::chr11:g.106100000::) Protein Fusion: mid-exon {PICALM:ALK}	NA	
EVT-P-0054253-T01-IM6-302	P-0054253-T01-IM6		PLEKHA7-ALK	PLEKHA7	unknown	three_prime	29448396	19	1035	False	retained	1.0	False	Protein kinase domain					PLEKHA7 (NM_175058) - ALK (NM_004304) fusion: t(2;11)(p23;p15.1)(chr2:g.29448396::chr11:g.103500000::) Protein Fusion: mid-exon {PLEKHA7:ALK} PLEKHA7 (NM_175058) - ALK (NM_004304) fusion: c.52+5708:PPP1CB_c.3172+886:ALKInv	NA	
EVT-P-0059884-T01-IM7-303	P-0059884-T01-IM7		PPP1CB-ALK	PPP1CB	in-frame	three_prime	29447441	19	1058	True	retained	1.0	False	Protein kinase domain					PPP1CB (NM_206876) - ALK (NM_004304) fusion: c.52+5708:PPP1CB_c.3172+886:ALKInv	NA	
EVT-P-0048893-T01-IM6-307	P-0048893-T01-IM6		SPTBN1-ALK	SPTBN1	unknown	three_prime	29448346	19	1051	False	retained	1.0	False	Protein kinase domain					SPTBN1 (NM_003128) - ALK (NM_004304) fusion: c.763+1517:SPTBN1_c.3153:ALKInv	NA	
EVT-P-0053963-T02-IM6-308	P-0053963-T02-IM6		SPTBN1-ALK	SPTBN1	unknown	three_prime	29448345	19	1052	False	retained	1.0	False	Protein kinase domain					SPTBN1 (NM_003128) - ALK (NM_004304) fusion: c.764-1123:SPTBN1_c.3154:ALKInv	NA	

3 Figures

ALK domain-retention track

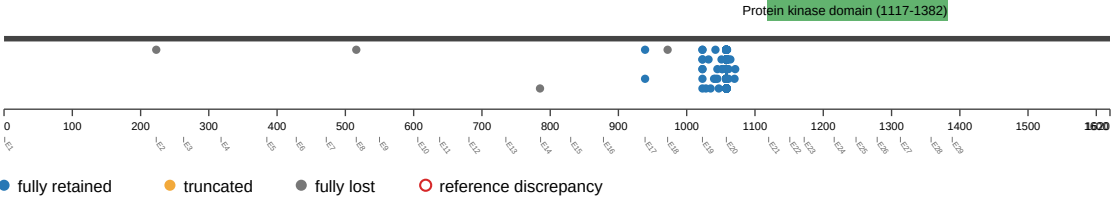
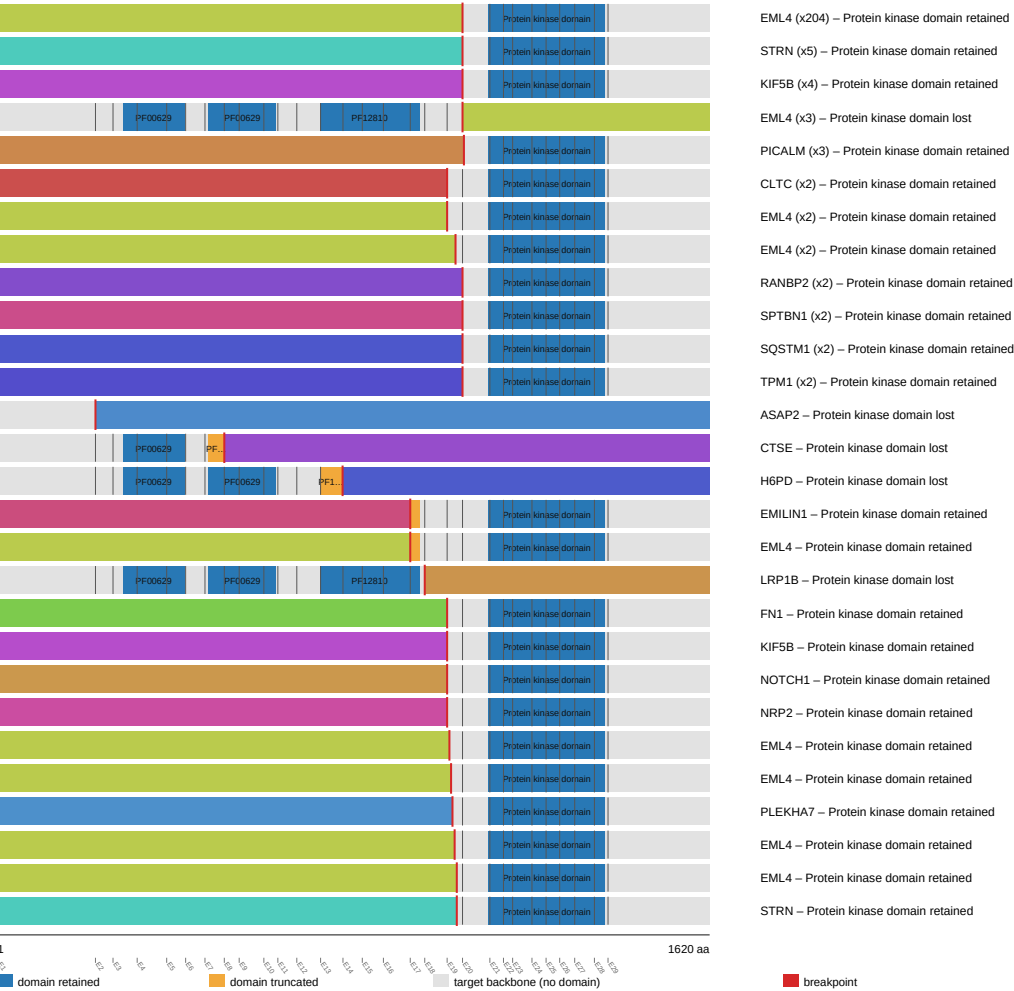


Figure 1: domain retention outliers

ALK fusion-transcript schematic

partner block → breakpoint → retained ALK portion (domain-colored, exon ticks)



Showing the top 28 of 50 partner/breakpoint groups by recurrence; see results.tsv for the complete list.

Figure 2: fusion schematic

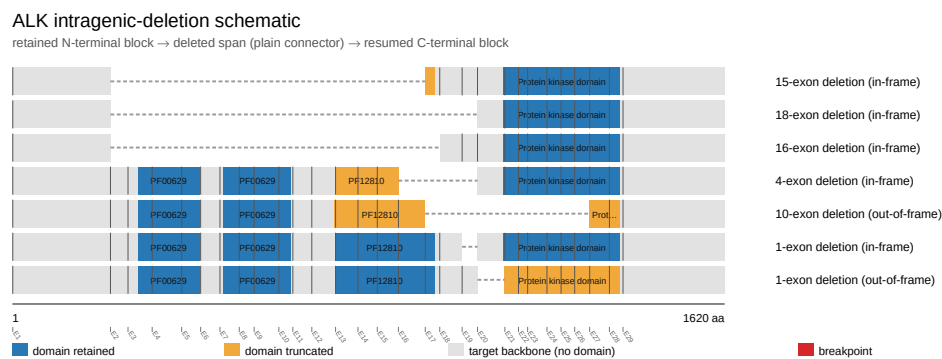


Figure 3: intragenic deletion schematic

Reference vs msk_impact_50k_2026



Figure 4: reference comparison